HP407 – EVIDENCE REVIEW AND SYNTHESIS FOR DECISION MAKING

SUMMATIVE ASSESSMENT 2022/23

CANDIDATE NUMBER:	**5 1 6 0 0**
ASSESSMENT TITLE	**Efficacy of caregiver interventions in low- and middle-income countries: a meta-analysis and systematic review**
WORD COUNT	

Please tick this box if you would be happy for your assessment to be shared with future cohorts: ☒

Efficacy of caregiver interventions in low- and middle-income countries (LMICs): a meta-analysis and systematic review

ABSTRACT

Background

More than two thirds of dementia patients live in LMICs, where the family is often the main caregiver and the support for caregivers are scarce. Distinct socioeconomic status and clinical profiles of dementia patients and their carers stresses the need to analyse the effectiveness of caregiver interventions in the LMIC settings.

Objective

To evaluate the effects of training and support programmes for caregivers of PwD on caregiver burden and quality of life in LMIC settings.

Design

Five reviewers screened title and abstracts. The risk of bias and data of each study was assessed by at least two reviewers, while disagreement was solved through group discussion among the reviewers.

Data sources

We further included 7 studies that focuses on support for caregivers published from 2019 to 2020 and updated the previous meta-analysis of 733 studies in LMICs, where 20 studies were found to have a significant proportion of caregiver interventions. The 27 studies, based on the type of measurements and data availability, are further curated. Finally, 14 and 12 studies are included for the meta-analysis of caregiver burden and caregiver quality of life, respectively.

Eligibility criteria

Only placebo controlled RCTs in English and Chinese were included. Included studies should be published between 2019 and 2020. The intervention under study was support or training programmes for caregivers and measured caregiver burden, caregiver health and/or care recipient health.

Results

Conclusion

TABLE OF CONTENTS

ABBREVIATIONS

BAI	Beck Anxiety Inventory
BDI	Beck Depression Inventory
CBI	Community-based intervention
CES-D 10	10-item Center for Epidemiologic Studies Depression Scale
CI	Confidential interval
CSI	Caregiver Strain Index
GHQ-28	28-item General Health Questionnaire
HIC	High-income country
IADL	Brody's Instrumental Activities of Daily Living Scale
LMIC	Low- and middle-income country
NPI-D	Score of distress of Neuropsychiatric Inventory
NPI-S	Score of severity of Neuropsychiatric Inventory
PHQ-4	Patient Health Questionnaire-4
PICOS	People, Intervention, Comparison, Outcome, Study design
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PwD	People with dementia
QoL	Quality of life
RCT	Randomized controlled trial
RoB	Risk of bias
SMD	Standardised mean difference
SD	Standardised deviation
SE	Standardised error
SF36	36-Item Short Form Survey
SQ	Spielberger's questionnaire
ZBI	Zarit Burden Interview

APPENDIX 1 PRISMA CHECKLIST

Section	Item #	Checklist item	Location
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	8
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	8
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	36
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	13
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	10
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	11
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	11
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	11

Section	Item #	Checklist item	Location
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	11
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	11
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	
Study characteristics	17	Cite each included study and present its characteristics.	
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	
	23b	Discuss any limitations of the evidence included in the review.	
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	
Competing interests	26	Declare any competing interests of review authors.	
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	

INTRODUCTION

Caregivers play a crucial role in providing support to individuals with dementia, but they often face significant challenges, especially high levels of stress and burden, making them the "invisible second patients" of dementia. Training and support interventions for caregivers have been proposed as a potential strategy to address these challenges and improve the quality of care for PwD.[1] However, the effectiveness of these interventions varies according to the accessibility of healthcare resources and the caregivers' knowledge and awareness of dementia care, which can affect their willingness to seek support.[2]

In LMICs, where more than two thirds of individuals with dementia live,[3] caregivers are often the main source of support, but support for caregivers is scarce.[4,5] Previous systematic reviews and meta-analyses have identified several types of interventions, such as psychoeducational interventions and multicomponent interventions, that can significantly alleviate caregiver burden.[6] However, these interventions are not all implemented and rarely validated in LMIC settings. Furthermore, a recent meta-analysis that includes studies from both LMICs and HICs suggests mixed evidence for coordinated community interventions.[7]

Given the unique clinical manifestations of dementia and inequalities in health resources and awareness of caregiver support in LMICs,[8] it is necessary to evaluate the effectiveness of caregiver interventions specifically in LMIC settings. This systematic review and meta-analysis aim to evaluate the effectiveness of training and support interventions for caregivers of people with dementia in LMICs. In this paper, we will also explore the potential moderators of the interventions, thus providing insights into effective caregiver support in dementia care.

METHODS

Identification and selection of studies

In addition to 733 records retrieved from the previous meta-analysis that we aim to update,[5] a database search of Ovid Medline, Ovid Embase, Ovid Global Health, Ovid

PsycINFO, EBSCO CINHL was performed to identify relevant studies published during 2019-2020 by a librarian on 21 September 20228, with the search strategy described in Appendix 1. The selection process is depicted in Figure 1. Rayyan was used for automated keyword extracting and highlighting and duplicate detection and removal.[9] Decisions were made through the consensus of all four reviewers. Inclusion criteria were as follows:

1) Publication type: original research published in English or Chinese
2) Population: PwD and their caregivers in LMICs
3) Intervention: support or training for caregiver
4) Comparators: No intervention as a control
5) Outcomes: the effect of caregiver support on patient and caregiver health

Data extraction

5 reviewers together identified the outcome measures to be extracted the data from the 7 newly included studies, with at least two reviewers recorded data items. One reviewer extracted the data from the 2019 meta-analysis using R,[10] with the detailed procedures described in Appendix 4. The extracted data items included publication details, country, randomisation methods, care setting, comparators, participant details, outcomes and risk of bias information. We did not extracted the source of funding data, comparing with the previous protocol.[5] Outcome measures related to the perceived burden, health and quality of life of dementia patients, or their caregivers were collected. Those not directly linked to the patient or caregiver health, such as measuring the knowledge and attitude of the caregiver, were discarded.

One study measured the subjective well-being and the positive experience of caregivers,[11] and another measured caregiver self-efficacy.[12] These outcome measures were not well-studied but closely associated with QoL.[13] Two studies only measured the mental and physical health perspectives of QoL.[14,15] Thus, although these data were collected, but they were insufficient to be meta-analysed.

Table 1 List of data items collected

Type of data items	Collected data items
Publication details	• Source: 2019 meta-analysis OR 2022 search • Authors • Year of publication • Title • Journal • Volume • Issue • Pages • DOI

Intervention details	• Intervention type: see Table 4 for details
	• Country: state where the research was conducted
	• Setting:
	  ○ urban OR rural
	  ○ nursing home OR community OR others
	• Study design
	  ○ parallel group OR cross-over
	  ○ randomised OR nonrandomised
	• Intervention and control names, as described in the paper
	• Detailed description of intervention quoted from the paper
	• Intervention duration described in the paper
	• Intervention duration converted to days
	• Intervention frequency

Participant details	• Participant inclusion criteria • Participant exclusion criteria • Type of dementia or mild cognitive impairment defined by the authors • Education • Gender
Outcome data	• List of health-related outcome measures for caregivers: see Table 4 for details • List of health-related outcome measures for care recipients: see Table 4 for details • Point of measurement • Baseline sample size • Baseline mean effect • Baseline SD • Sample size at the point of measurement • Mean effect at the point of measurement • SD of the effect at the point of measurement

Table 2. Summary of outcome measures

	Caregiver					Care recipient			
	Perceived burden	Quality of life	Depression	Anxiety	Neuropsychiatric symptoms	Cognitive function	Quality of life	Activity level	Neuropsychiatric symptoms
Amit 2008	ZBI								
Chen 2020						MoCA		BADL	BEHAVE-AD
Ghaffari 2019			GHQ-28	GHQ-28	GHQ-28				
Govindakumari 2020	Unclear						Unclear		
Guerra 2011	ZBI	WHO-QoL-Bref			NPI-D/NPI-S/SRQ-20		DEMQOL		
Hinton 2020	ZBI				PHQ-4				
JIANG 2012		WHO-QoL-Bref				MMSE			
Lin 2017	CBI								
Lök 2017	ZBI								

Mahdavi 2017	CSI								
Mercedeh 2014	CBI	SF-36	Ham-D						
Orawan 2018									
Pahlavanzadeh 2010	ZBI								
Pan 2019			CES-D						
Salamizadeh 2017									
Shata 2017	ZBI								
Söylemez 2016	CBI	WHO-QoL-Bref							
SU 2012						MMSE			
SUN 2010							SF-36		
TAN 2010						MMSE			
Uyar 2019	ZBI	SF-36	BDI	BAI	NPI-D		QoL-AD		NPI-S
Wang 2010b						MMSE		IADL	
Wang 2012e		WHO-QoL-Bref							
Wang 2014a						MMSE		IADL	
Yang 2017a						MMSE			
Zarepour 2020				SQ					
ZHAO 2010							SF-36		

Risk of bias

Risk of bias assessment was performed as guided with RoB2 tool, the revised Cochrane risk-of-bias tool for randomized trials,[16] and visualised with the R package "robvis".[17] Six included studies were assessed by two reviewers and the remaining one was assessed by all four reviewers, with the disagreement settled through group consensus. The risk of bias of the 22 previous studies was obtained from the records of the 2019 meta-analysis. Risk of bias across studies was examined with a funnel plot for each meta-analysis. Trim-and-fill was used to correct for publication bias.

Qualitative synthesis

The included studies should be placebo-control parallel-group RCT. No cross-over studies were included to avoid complexity. The participants should be the informal caregivers of dementia patients. The interventions involved should be further categorised based on the definitions by Walter and Pinquart (2020) and their previous work (summarised in Table 3) and will be analysed in subgroups accordingly.[18,19,19] The risk of bias assessed with RoB 2 should provide further information about the trial. We performed meta-analysis only when there were at least three studies measuring the same outcomes, or else we would considering the information insufficient as it was impossible to conduct further analyses such as trim-and-fill.

Table 3 Definitions of types of caregiver support

Type	Definition
Psychoeducational interventions (PEI)	Structured educational programme in which the interventionists provide information about dementia and AD and train caregivers to effectively deal with the disease-related problems of care recipients. Additional support may be provided, but it needs to be secondary to the educational content.
Cognitive-behavioural therapy (CBT)	Structured or unstructured programmes involving dentification and modification of the beliefs underlying caregivers or other techniques from CBT, to develop new behavioural pattern for caregiving
Counselling/case management (CCM)	Individualised advisory services and support, aiming to help caregivers solve their specific personal needs not necessarily related to caregiving
General support (GS)	Peer- or professional-led groups to enable discussion and information sharing about dementia- and caregiving-related issues and to allow people to share their personal feelings
Respite	Provision of substitute care upon caregivers' request or when caregivers are in need, thus allowing caregivers to temporarily exempted from care responsibility
Cognitive and functional training (CFT)	Memory clinics or training programmes aimed at improvement of caregivers' cognitive and functional impairments, rather than care recipients'
Multi-component intervention (MCI)	A combination of above-mentioned types of interventions. Further information must be provided about the intervention types of each component of the intervention.
Miscellaneous	None of above-mentioned types

Statistical analysis

Change scores and its SMD were calculated as per Morris (2008),[20] and estimated with Hedge's *g*,[21] suing R packages 'metafor'. We used random-effects models for meta-analysis, as we expected high heterogeneity across studies, due to their diversity of methods of intervention and outcome measurement.[22–24] Their heterogeneous, as measured with I2, were reported for each meta-analysis. Sensitivity analyses of different measurement time points and measurement scales were conducted to examine the robustness of results.[25,26] In light of the heterogeneity of the studies, we used prediction interval to identify the range of effect size where future studies might fall, in addition to confidence intervals that summarised our analysis.

RESULTS

Identified and eligible studies

Deviating from the original protocol,[5] we only included RCTs because it is considered the most reliable evidence on the effectiveness of an intervention due to its scientifically rigorous methods, including but not limited randomisation, minimising the risk of the confounding factors affecting the results.[27] The same criteria were applied to selection of the records from the 2019 meta-analysis. Ultimately, 7 studies from the 2022 search and 20 studies from the 2019 meta-analysis were included.

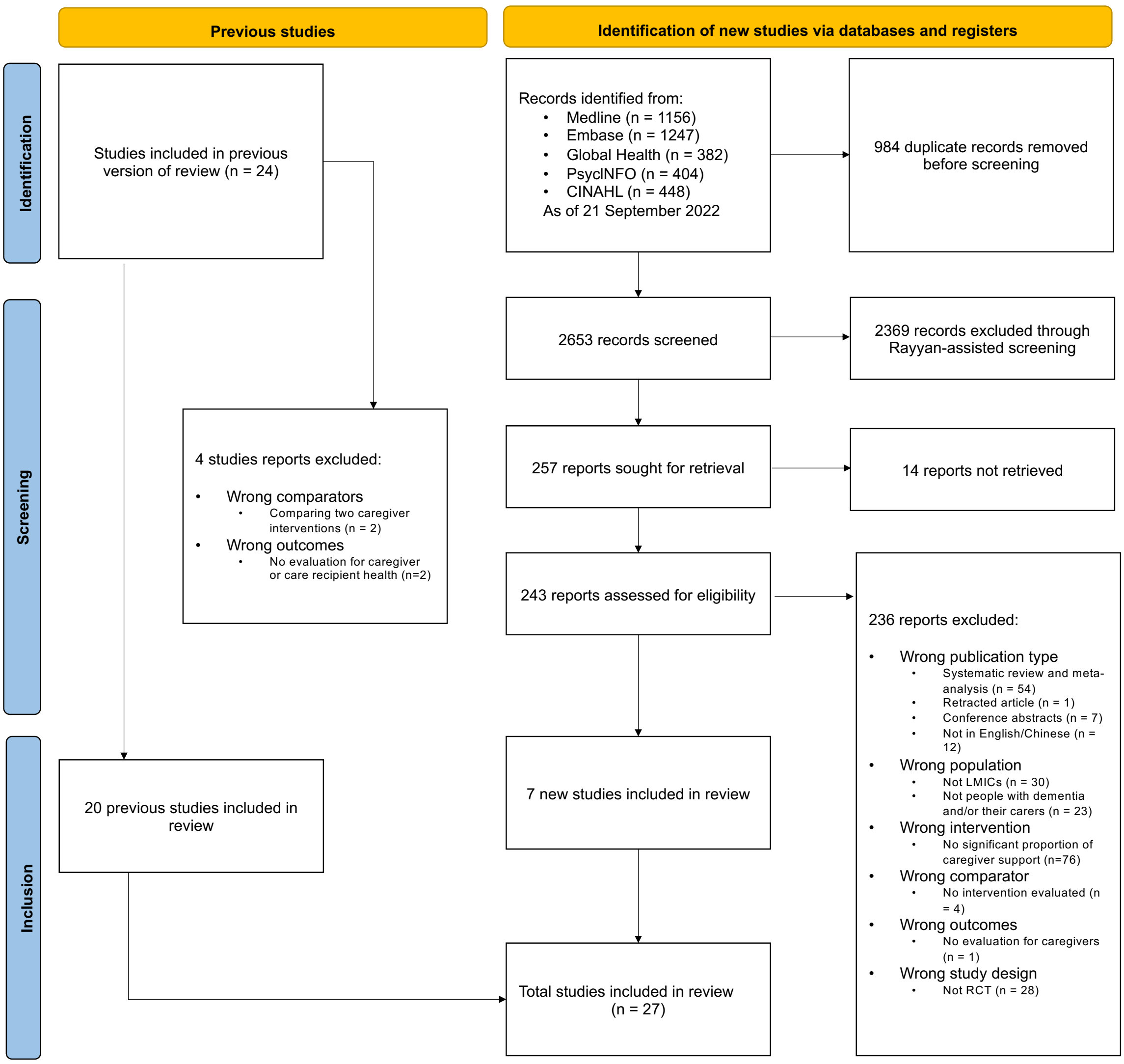


Figure 1. Flow chart of paper screening (PRISMA)

Characteristics of included studies

Of 27 trials, 12 were based in China, 6 in Iran, 3 in Turkey, 2 in India, and 1 each in Egypt, Peru, Thailand and Vietnam, with the date of publication ranging from 2008 to 2020 (Figure 2). The publication dates of these trials ranged from 2008 to 2020, as shown in Figure 3. Most of these trials used individual randomization, with only two studies randomizing subjects based on the community. All of the included research studies focused on dementia patients and their caregivers, who were similar in terms of education and gender. Specifically, more than 70% of caregivers were female and few had received college education, a finding that is consistent with previous research. However, there were few studies conducted in rural settings or in low-income countries except India and Vietnam.

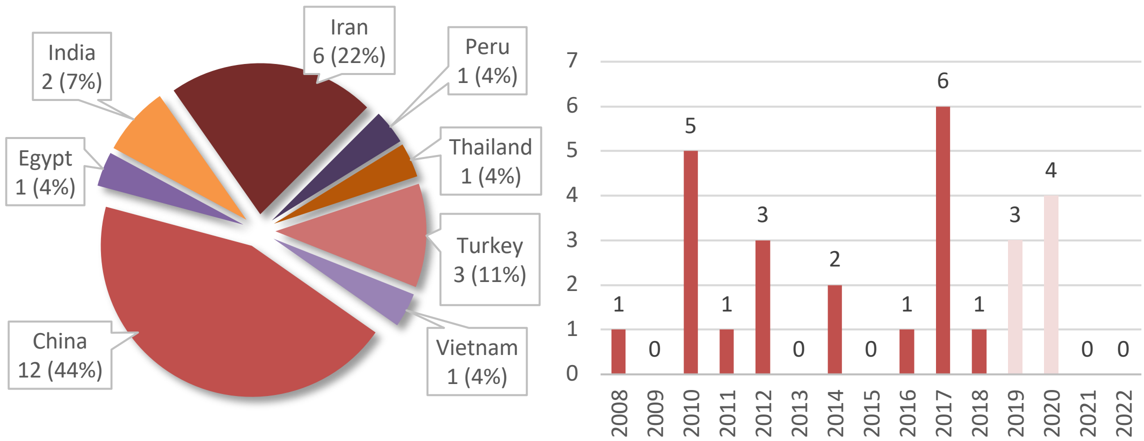


Figure 2. Location and publication date of included studies

There was significant diversity in the content and duration of the interventions used in these trials. 16 of the trials were categorized as multicomponent interventions (MCIs), 4 were cognitive-behavioural therapies (CBTs), 2 were miscellaneous interventions (related to spirituality), 3 were psychosocial interventions (PEIs), and 1 each was a group support (GS) intervention and a caregiver-focused therapy (CFT). This suggests that MCIs are more popular in low- and middle-income countries (LMICs) than expected. Among the 16 MCIs, PEIs were the most common component, used in 15 MCIs, followed by case management (CCM) in 12 MCIs and GS in 11 MCIs. However, it is important to note that at least five trials in China[a] used the same MCI pattern of PEI+CCM+GS, all of which lasted for one year and only measured outcomes for care recipients. The other included studies were more diverse in terms of interventions and outcome measurement.

Table 4 Summary statistics of types of caregiver support

Table 4.1 Summary of types of caregiver support

Type	# of studies	Study size	Length of intervention in days
		Median (Min; Max)	
PEI	3	50 (40; 70)	35 (28; 49)
CBT	4	63 (20; 99)	40 (35; 150)
CCM	N/A	N/A	N/A
GS	1	102	56
Respite	N/A	N/A	N/A
CFT	1	99	30
MCI	16	64 (38; 141)	174 (14; 365)
MISC	2	72 (54; 89)	35

Table 4.2 Components of MCIs

	PEI	CBT	GS	CFT	CCM
Amit 2008	✓		✓		✓
Guerra 2011	✓	✓			
Hinton 2020	✓	✓		✓	✓
JIANG 2012	✓				✓
Lin 2017	✓		✓		✓
Shata 2017	✓	✓	✓		
Söylemez 2016	✓		✓		✓
SU 2012	✓				✓
SUN 2010	✓		✓		✓
TAN 2010	✓		✓		✓
Uyar 2019	✓		✓		
Wang 2010b	✓		✓		✓
Wang 2012e			✓		✓
Wang 2014a	✓		✓		✓
Yang 2017a	✓		✓		✓
ZHAO 2010	✓		✓		✓

Caregiver outcomes were measured in 18 trials, with 12 measuring caregiver outcomes, including perceived burden, quality of life, depression, neuropsychiatric symptoms, and caregiver anxiety (Table 2.1). Care recipient outcomes were measured in 12 trials, including cognitive function, quality of life, activity level, and neuropsychiatric symptoms (Table 2.2). The measurement methods used in these studies differed. For example, the Zarit Burden Interview (ZBI), Caregiver Burden Inventory (CBI), and Caregiver Strain Index (CSI) were all used to measure caregiver strain. Thus, we will convert them to Hedge's g, a bias-corrected SMD estimator. Most

[a] SUN 2010, Wang 2010b, Wang 2014a, Yang 2017a, ZHAO 2010

studies only involved group-wise comparison and did not report effect sizes, although positive effects of the interventions on various outcomes were typically described, except that Uyar (2019) oddly reported decreased caregiver QoL after intervention.

Table 5 Summary of effect sizes

Outcome measures	Estimation	SE	95% CI		Prediction Interval		I^2%
			Lower	Upper	Lower	Upper	
Caregiver burden	-1.74*	0.54	-2.80	-0.67	-5.49	2.02	96.98
Caregiver depression	-1.28**	0.26	-1.79	-0.77	-2.22	-0.34	61.96
Caregiver distress	-0.42	0.27	-0.95	0.11	-0.95	0.11	0.00
Caregiver mental health	-1.89	0.98	-3.81	0.03	-5.15	1.37	93.30
Caregiver mental QoL	0.56**	0.27	0.03	1.10	0.03	1.10	0.00
Caregiver physical QoL	0.78**	0.28	0.24	1.33	0.24	1.33	0.00
Caregiver QoL	1.46*	0.37	0.74	2.18	-0.21	3.13	84.03
Patient activities of daily life	1.15*	0.44	0.29	2.00	-0.26	2.55	86.10
Patient cognitive function	0.57**	0.08	0.40	0.74	0.40	0.74	0.00
Patient symptoms	-0.72	0.52	-1.75	0.31	-2.43	1.00	88.98
Patient quality of life	1.60	0.95	-0.26	3.46	-2.71	5.91	97.07

* significant according to 95% CI; ** significant according to both 95% CI and prediction interval; only random-effects models are used regardless of heterogeneity, which yield more conservative estimation than fixed-effect model; heterogeneity

Risk of bias within studies

The risk of bias is summarised in Figure 3, with the details listed in 0. Only one study was identified to be at low risk of bias, while the rest all at high risk of bias or related to some concerns. We also identified two with flawed statistical analysis, yet this affected only the authors' interpretation of the outcomes but not the outcomes. Thus, the outcome data of these studies were also included.

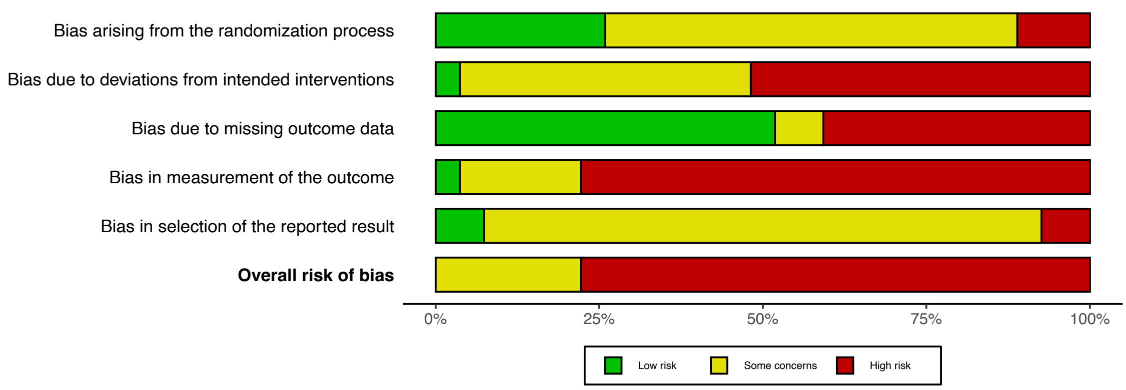


Figure 3.1 Risk of bias summary of 27 included studies

Risk of bias domains

Study	D1	D2	D3	D4	D5	Overall
Chen 2020	+	−	+	X	−	X
Ghaffari 2019	X	X	X	X	−	X
Govindakumari 2020	−	X	X	X	−	X
Hinton 2020	+	−	−	+	+	−
Pan 2019	+	X	X	X	−	X
Uyar 2019	X	X	+	X	X	X
Zarepour 2020	+	+	+	X	+	X

Figure 3.2 Risk of bias summary of 7 studies from the 2020 search results

Risk of bias domains

Study	D1	D2	D3	D4	D5	Overall
Amit 2008	+	−	+	X	−	X
Guerra 2011	+	−	+	X	−	X
JIANG 2012	−	−	+	−	−	−
Lin 2017	−	−	+	−	−	−
Lök 2017	−	X	X	X	−	X
Mahdavi 2017	−	X	X	X	X	X
Mercedeh 2014	X	X	X	X	−	X
Orawan 2018	−	−	+	X	−	X
Pahlavanzadeh 2010	−	X	X	X	−	X
Salamizadeh 2017	−	X	X	X	−	X
Shata 2017	+	X	X	X	−	X
Söylemez 2016	−	−	−	X	−	X
SU 2012	−	−	+	X	−	X
SUN 2010	−	−	+	−	−	−
TAN 2010	−	−	+	−	−	−
Wang 2010b	−	X	+	X	−	X
Wang 2012e	−	X	X	X	−	X
Wang 2014a	−	X	+	X	−	X
Yang 2017a	−	X	X	X	−	X
ZHAO 2010	−	−	+	−	−	−

Domains:
D1: Bias arising from the randomization process.
D2: Bias due to deviations from intended intervention.
D3: Bias due to missing outcome data.
D4: Bias in measurement of the outcome.
D5: Bias in selection of the reported result.

Judgement
X High
− Some concerns
+ Low

Figure 3.3 Risk of bias summary of 20 studies from the 2019 meta-analysis

Figure 3 Risk of bias summary of 27 included studies

Risk of bias across studies

Qualitative review of studies

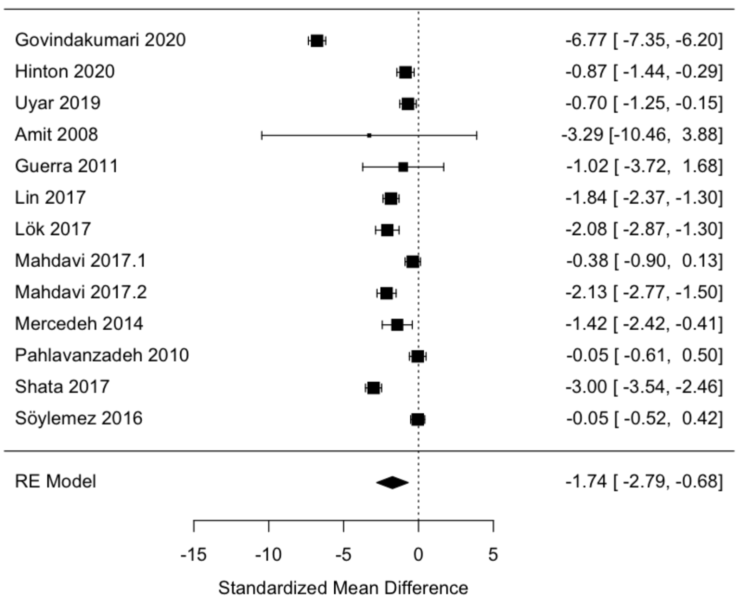
Meta-analysis results

Additional analyses

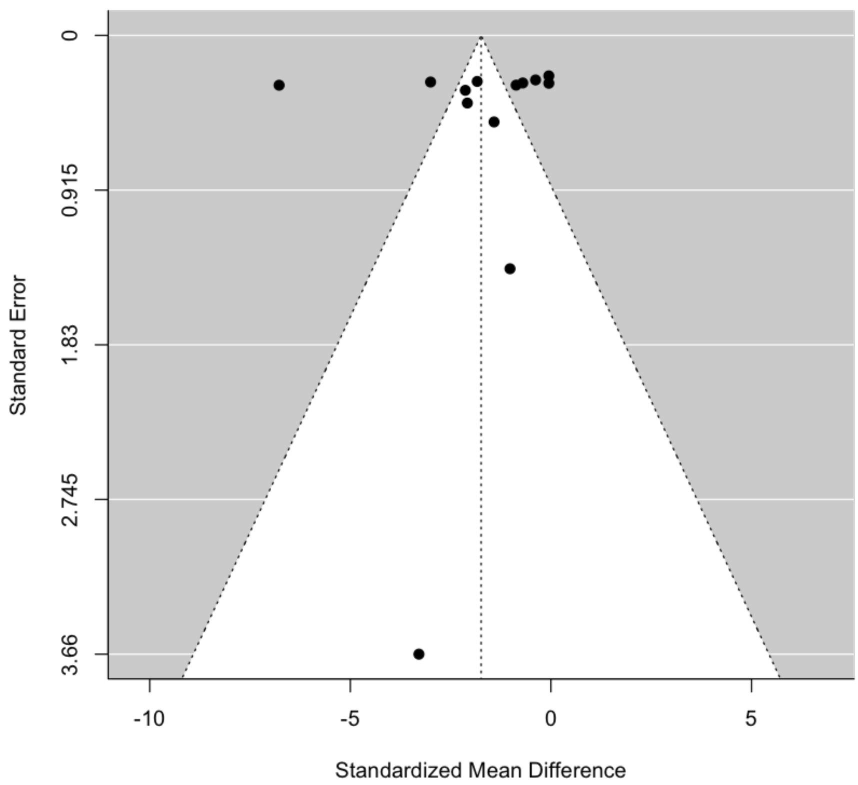
DISCUSSION

Statement of principal findings

Our meta-analysis shows that caregiver interventions led to significant improvement in caregiver anxiety, caregiver depression, caregiver psychological quality of life, caregiver physical quality of life, caregiver overall quality of life, care recipient cognitive function. However, considering the heterogeneity of the studies into consideration, the improvement in caregiver burden, caregiver QoL and caregiver activities of daily life is controversial as indicated by their prediction intervals. The analysis shows moderate to large effect sizes for most outcomes, yet

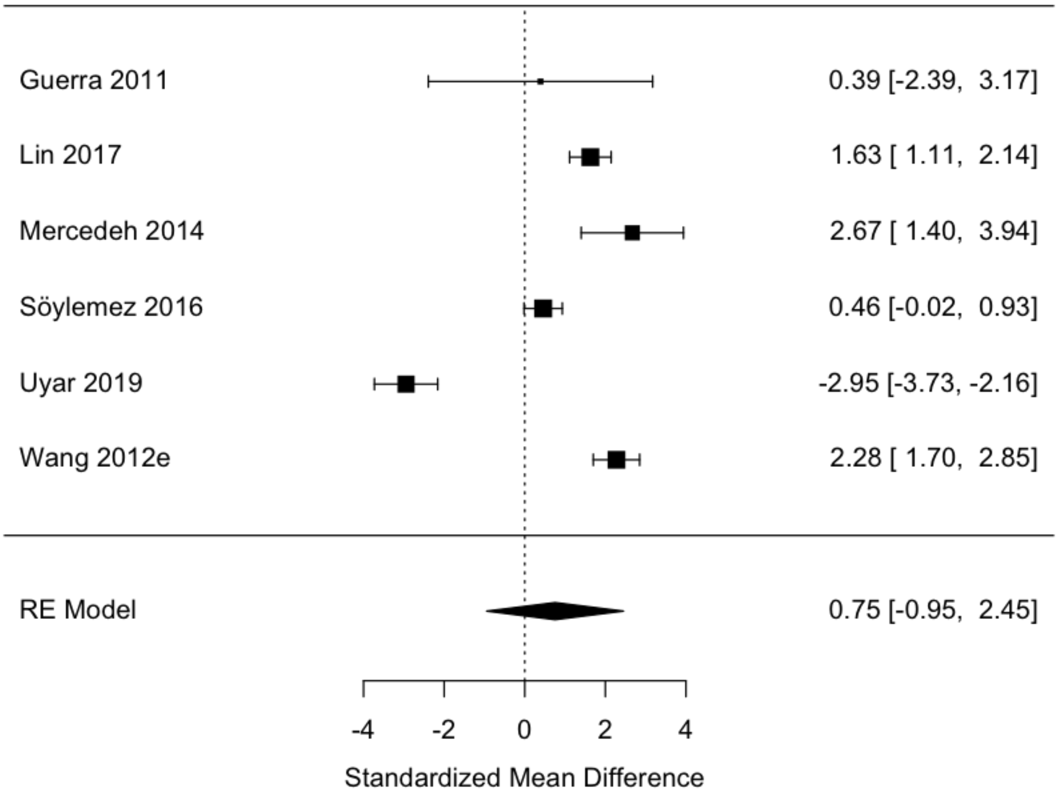



```
Random-Effects Model (k = 13; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 3.3344 (SE = 1.4995)
tau (square root of estimated tau^2 value):      1.8260
I^2 (total heterogeneity / total variability):   96.95%
H^2 (total variability / sampling variability):  32.80

Test for Heterogeneity:
Q(df = 12) = 449.5197, p-val < .0001

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -1.7393  0.5383  -3.2312  0.0012  -2.7943  -0.6843  **

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

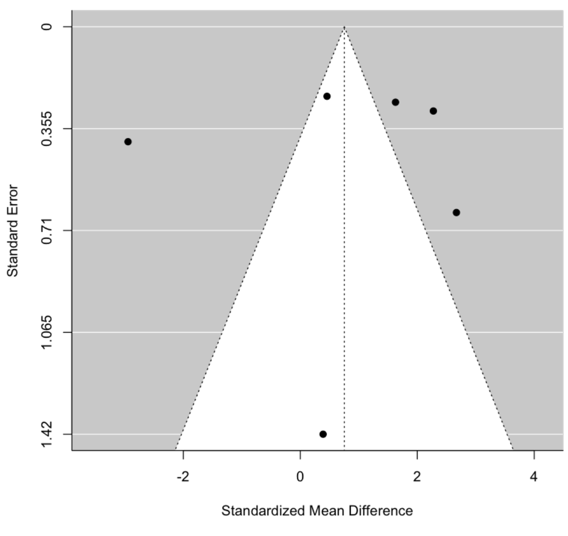



```
Estimated number of missing studies on the right side: 0 (SE = 2.0265)

Random-Effects Model (k = 13; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 3.3344 (SE = 1.4995)
tau (square root of estimated tau^2 value):      1.8260
I^2 (total heterogeneity / total variability):   96.95%
H^2 (total variability / sampling variability):  32.80

Test for Heterogeneity:
Q(df = 12) = 449.5197, p-val < .0001

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -1.7393  0.5383  -3.2312  0.0012  -2.7943  -0.6843  **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Quality of life

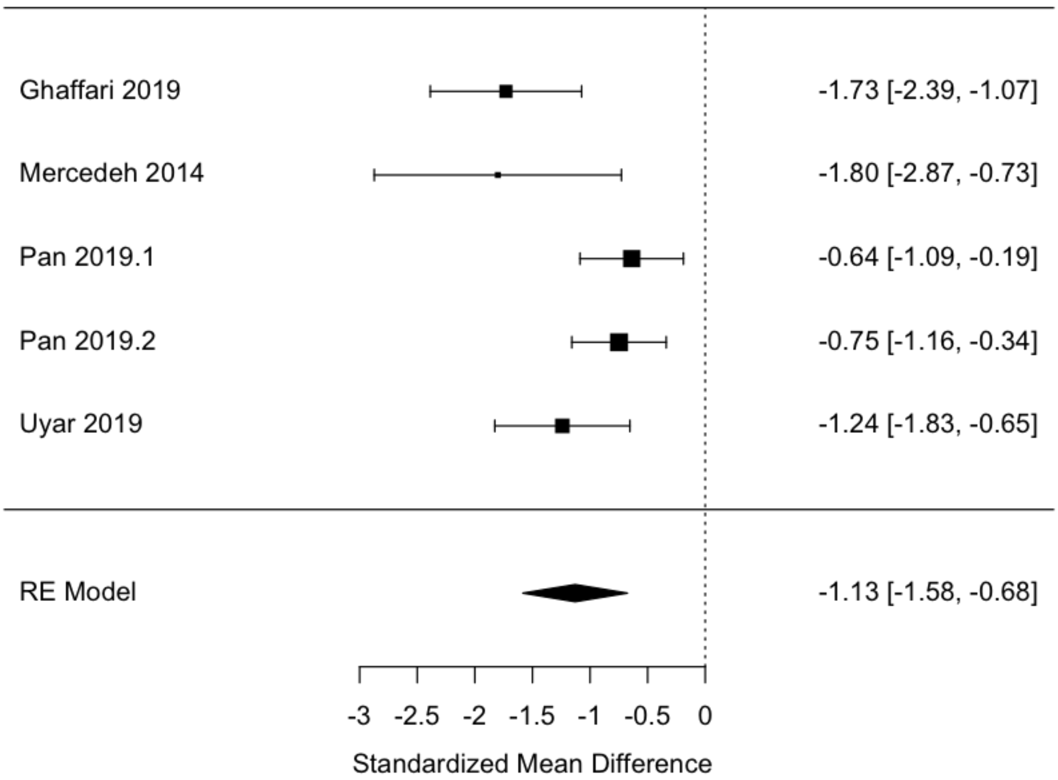


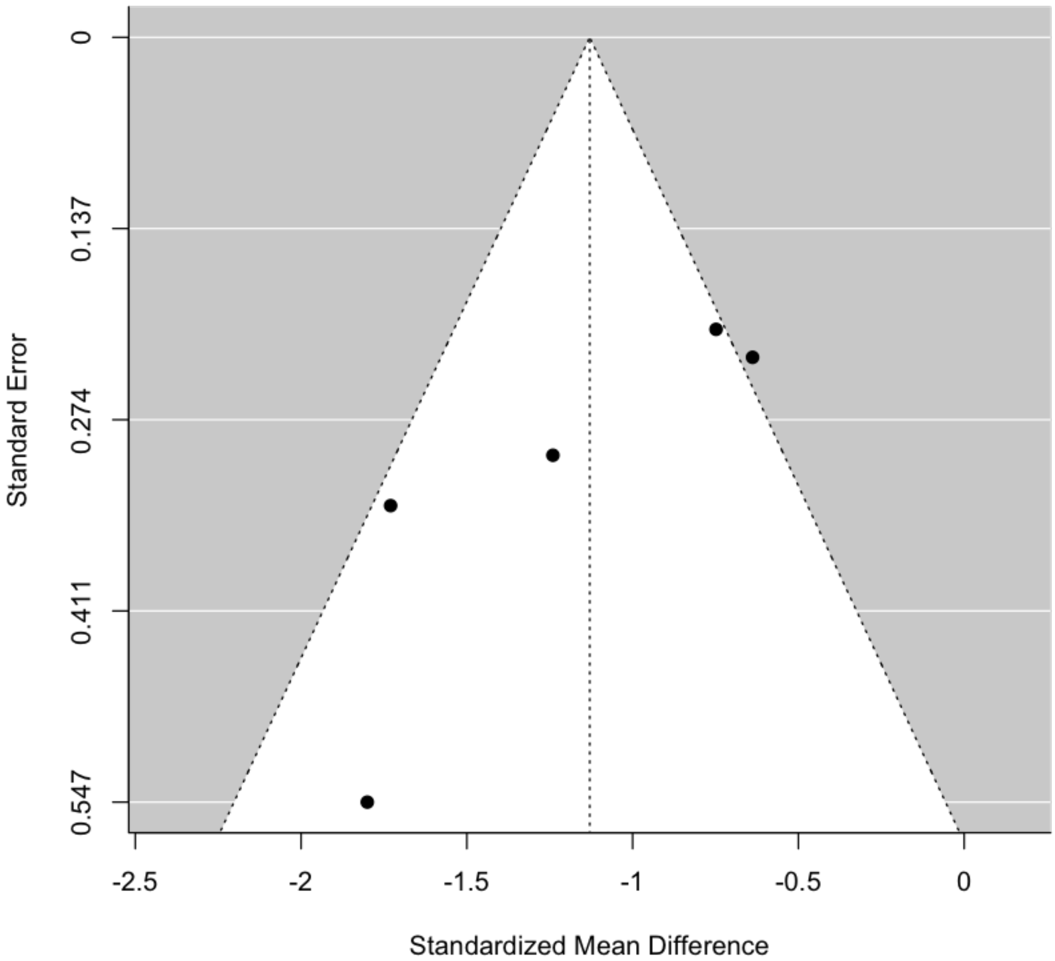
```
Random-Effects Model (k = 6; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 4.1233 (SE = 2.8394)
tau (square root of estimated tau^2 value):      2.0306
I^2 (total heterogeneity / total variability):   97.00%
H^2 (total variability / sampling variability):  33.34

Test for Heterogeneity:
Q(df = 5) = 132.6997, p-val < .0001

Model Results:

estimate      se    zval    pval    ci.lb   ci.ub
  0.7523  0.8666  0.8682  0.3853  -0.9461  2.4508

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```



```
Estimated number of missing studies on the right side: 0 (SE = 1.5783)

Random-Effects Model (k = 6; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 4.1233 (SE = 2.8394)
tau (square root of estimated tau^2 value):      2.0306
I^2 (total heterogeneity / total variability):   97.00%
H^2 (total variability / sampling variability):  33.34

Test for Heterogeneity:
Q(df = 5) = 132.6997, p-val < .0001

Model Results:

estimate      se    zval    pval    ci.lb   ci.ub
  0.7523  0.8666  0.8682  0.3853  -0.9461  2.4508

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Caregiver depression


Ghaffari 2019
Mercedeh 2014
Pan 2019.1
Pan 2019.2
Uyar 2019
RE Model
-1.73 [-2.39, -1.07]
-1.80 [-2.87, -0.73]
-0.64 [-1.09, -0.19]
-0.75 [-1.16, -0.34]
-1.24 [-1.83, -0.65]
-1.13 [-1.58, -0.68]
-3 -2.5 -2 -1.5 -1 -0.5 0
Standardized Mean Difference


Standard Error
0
0.137
0.274
0.411
0.547
-2.5
-2
-1.5
-1
-0.5
0
Standardized Mean Difference

```
Estimated number of missing studies on the right side: 1 (SE = 1.7009)

Random-Effects Model (k = 6; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 0.1680 (SE = 0.1773)
tau (square root of estimated tau^2 value):      0.4099
I^2 (total heterogeneity / total variability):   63.82%
H^2 (total variability / sampling variability):  2.76

Test for Heterogeneity:
Q(df = 5) = 13.0544, p-val = 0.0229

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -1.0418  0.2183  -4.7729  <.0001  -1.4696  -0.6140  ***

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Random-Effects Model (k = 5; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 0.1711 (SE = 0.1879)
tau (square root of estimated tau^2 value):      0.4136
I^2 (total heterogeneity / total variability):   67.47%
H^2 (total variability / sampling variability):  3.07

Test for Heterogeneity:
Q(df = 4) = 11.4619, p-val = 0.0218

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -1.1293  0.2317  -4.8731  <.0001  -1.5835  -0.6751  ***

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Strengths and weaknesses of the study

This study is the only meta-analysis specifically focused on caregiver interventions in low- and middle-income countries (LMICs). However, it is important to note that most of the studies included in this analysis have a high risk of bias, despite being randomized controlled trials (RCTs). High-quality evidence with low risk of bias is still scarce in LMICs. As a result, our synthesized evidence may not be fully conclusive due to the limited number and quality of the studies included.

A further limitation of this study is the language representation of the included studies. During the initial search, we included 17 studies that were not in English, but excluded 12 papers written in other languages that we were unable to read. Among the excluded papers, we identified one relevant study written in Farsi that we were unable to retrieve data from due to language barriers. This study should be included in future research.

Additionally, this study used summary statistics (mean and standard deviation) rather than original patient-level data to estimate between-group standardized mean differences (SMD). While this approach is commonly used, it has been found to be controversial as it can lead to bias due to issues such as incorrect estimation of correlation coefficients and statistical dependency.[28,29] Therefore, it would be beneficial to acquire patient-level data for future analysis.

Strengths and weaknesses in relation to other studies

Meaning of the study

Unanswered questions and future research

CONCLUSIONS

REFERENCES

1. Giebel C. "Current dementia care: what are the difficulties and how can we advance care globally?" *BMC Health Services Research*. 2020;20(1):414. doi:10.1186/s12913-020-05307-1

2. Zwingmann I, Dreier-Wolfgramm A, Esser A, et al. Why do family dementia caregivers reject caregiver support services? Analyzing types of rejection and associated health-impairments in a cluster-randomized controlled intervention trial. *BMC Health Serv Res*. 2020;20(1):121. doi:10.1186/s12913-020-4970-8

3. Mattap SM, Mohan D, McGrattan AM, et al. The economic burden of dementia in low- and middle-income countries (LMICs): a systematic review. *BMJ Glob Health*. 2022;7(4):e007409. doi:10.1136/bmjgh-2021-007409

4. Fam J, Mahendran R, Kua EH. Dementia care in low and middle-income countries. *Current opinion in psychiatry*. 2019;32(5):461-464. doi:10.1097/yco.0000000000000523

5. Salcher-Konrad M, Naci H, McDaid D, et al. Effectiveness of interventions for dementia in low- and middle-income countries: protocol for a systematic review, pairwise and network meta-analysis. *BMJ Open*. 2019;9(6):e027851. doi:10.1136/bmjopen-2018-027851

6. Walter E, Pinquart M. How Effective Are Dementia Caregiver Interventions? An Updated Comprehensive Meta-Analysis. Heyn PC, ed. *The Gerontologist*. 2020;60(8):e609-e619. doi:10.1093/geront/gnz118

7. Backhouse A, Ukoumunne OC, Richards DA, McCabe R, Watkins R, Dickens C. The effectiveness of community-based coordinating interventions in dementia care: a meta-analysis and subgroup analysis of intervention components. *BMC Health Serv Res*. 2017;17:717. doi:10.1186/s12913-017-2677-2

8. Alladi S, Hachinski V. World dementia: One approach does not fit all. *Neurology*. 2018;91(6):264-270. doi:10.1212/wnl.0000000000005941

9. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan—a web and mobile app for systematic reviews. *Syst Rev*. 2016;5(1):210. doi:10.1186/s13643-016-0384-4

10. R Core Team. R: A language and environment for statistical computing. Published online October 31, 2022. https://www.R-project.org/

11. Pankong O, Pothiban L, Sucamvang K, Khampolsiri T. A randomized controlled trial of enhancing positive aspects of caregiving in Thai dementia caregivers for dementia. *Pacific Rim International Journal of Nursing Research*. 2018;22(2):131-143. https://he02.tci-thaijo.org/index.php/PRIJNR/article/view/86178

12. Salamizadeh A, Mirzaei T, Ravari A. The impact of spiritual care education on the self-efficacy of the family caregivers of elderly people with Alzheimer's disease. *International journal of community based nursing and midwifery*. 2017;5(3):231. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5478743/

13. Martyr A, Nelis SM, Quinn C, et al. Living well with dementia: a systematic review and correlational meta-analysis of factors associated with quality of life, well-being and life satisfaction in people with dementia. *Psychological Medicine*. 2018;48(13):2130-2139. doi:10.1017/S0033291718000405

14. Uyar F, Özmen D, Mavioğlu H, Atalay N. Assessment of the impact of dementia care and support program in both patient and caregiver outcomes: an intervention study. *Türk Geriatri Dergisi*. 2019;22(4):482-493. doi:10.31086/tjgeri.2020.127

15. Guerra M, Ferri CP, Fonseca M, Banerjee S, Prince M. Helping carers to care: the 10/66 dementia research group's randomized control trial of a caregiver intervention in Peru. *Rev Bras Psiquiatr*. 2010;33(1):47-54. doi:10/bmn2ch

16. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366. doi:10.1136/bmj.l4898

17. McGuinness LA, Higgins JPT. Risk-of-bias VISualization (robvis): An R package and Shiny web app for visualizing risk-of-bias assessments. *Res Syn Meth*. 2021;12(1):55-61. doi:10.1002/jrsm.1411

18. Sörensen S, Pinquart M, Duberstein P. How Effective Are Interventions With Caregivers? An Updated Meta-Analysis. *The Gerontologist*. 2002;42(3):356-372. doi:10.1093/geront/42.3.356

19. Pinquart M, Sörensen S. Helping caregivers of persons with dementia: which interventions work and how large are their effects? *International Psychogeriatrics*. 2006;18(4):577-595. doi:10.1017/S1041610206003462

20. Morris SB. Estimating Effect Sizes From Pretest-Posttest-Control Group Designs. *Organizational Research Methods*. 2008;11(2):364-386. doi:10.1177/1094428106291059

21. Viechtbauer W. Conducting Meta-Analyses in R with the metafor Package. *Journal of Statistical Software*. 2010;36:1-48. doi:10.18637/jss.v036.i03

22. Harrer M, Cuijpers P, Furukawa TA, Ebert DD. *Doing Meta-Analysis With R: A Hands-On Guide*. 1st ed. Chapman & Hall/CRC Press; 2021. https://www.routledge.com/Doing-Meta-Analysis-with-R-A-Hands-On-Guide/Harrer-Cuijpers-Furukawa-Ebert/p/book/9780367610074

23. Guido Schwarzer, James R. Carpenter, Gerta Rücker. *Meta-Analysis with R*. Springer International Publishing; 2015. Accessed December 12, 2022. https://link.springer.com/book/10.1007/978-3-319-21416-0

24. Chen DG (Din), Peace KE. *Applied Meta-Analysis with R*. CRC Press; 2013.

25. Fu R, Holmer HK. *Change Score or Followup Score? An Empirical Evaluation of the Impact of Choice of Mean Difference Estimates*. Agency for Healthcare Research and Quality (US); 2015. Accessed December 13, 2022. http://www.ncbi.nlm.nih.gov/books/NBK285983/

26. Higgins J, Thomas J, Chandler J, et al., eds. 6.5.2.8 Imputing standard deviations for changes from baseline. In: *Cochrane Handbook for Systematic Reviews of Interventions*. Second Edition. Cochrane book series. The Cochrane Collaboration and John Wiley & Sons Ltd.; 2019. https://lccn.loc.gov/2019024539

27. Akobeng AK. Understanding randomised controlled trials. *Archives of Disease in Childhood*. 2005;90(8):840-844. doi:10.1136/adc.2004.058222

28. Cuijpers P, Weitz E, Cristea IA, Twisk J. Pre-post effect sizes should be avoided in meta-analyses. *Epidemiol Psychiatr Sci*. 2016;26(4):364-368. doi:10.1017/S2045796016000809

29. Kösters M. Every effect size has its place: a commentary on the avoidance of pre–post effect sizes. *Epidemiol Psychiatr Sci*. 2017;26(4):369-370. doi:10.1017/S204579601700004X

30. Chen M, Bai Y, Zhou X, Chen W, He D, Li Y. Effect of Extended Nursing on the Behavioral and Psychological Symptoms and Cognitive Dysfunction of Patients with Moderate and Severe Alzheimer's Disease. *ijps*. 2020;82. doi:10.36468/pharmaceutical-sciences.spl.137

31. Govindakumari R, Goothy SSK, Raghvan V. Effectiveness of home-based cognitive training program on a cognitive level, the activity of daily living among elderly persons with mild cognitive impairment in selected community settings of Puducherry. *International Journal of Research in Pharmaceutical Sciences*. 2020;11(4):5301-5309. doi:10.26452/ijrps.v11i4.3149

32. Ghaffari F, Rostami M, Fotokian Z, Hajiahmadi M. Effectiveness of Resilience Education in the Mental Health of Family Caregivers of Elderly Patients with Alzheimer's Disease. *Iran J Psychiatry Behav Sci*. 2019;13(3). doi:10.5812/ijpbs.69507

33. Hinton L, Nguyen H, Nguyen HT, et al. Advancing family dementia caregiver interventions in low- and middle-income countries: A pilot cluster randomized controlled trial of Resources for Advancing Alzheimer's Caregiver Health in Vietnam (REACH VN). *A&D Transl Res & Clin Interv*. 2020;6(1). doi:10.1002/trc2.12063

34. Pan Y, Chen R, Yang D. The Role of Mutuality and Coping in a Nurse-Led Cognitive Behavioral Intervention on Depressive Symptoms Among Dementia

Caregivers. *Research in Gerontological Nursing*. 2019;12(1):44-55. doi:10.3928/19404921-20181212-01

35. Zarepour A, Hazrati M, Kadivar AA. The Impact of Educational Intervention on the Anxiety of Family Caregivers of the Elderly with Dementia: A Randomized Controlled Tri. *IJCBNM*. 2020;8(3). doi:10.30476/ijcbnm.2020.81680.0

36. Dias A, Dewey ME, D'Souza J, et al. The Effectiveness of a Home Care Program for Supporting Caregivers of Persons with Dementia in Developing Countries: A Randomised Controlled Trial from Goa, India. Brayne C, ed. *PLoS ONE*. 2008;3(6):e2333. doi:10.1371/journal.pone.0002333

37. Jiang G xuan, Li M qiu, Huang H hua, Chai Z fang, Wu A lian, Zhang G sheng. The curative effect of whole course rehabilitation nursing intervention on mild-to-moderate Alzheimer's disease patients. *Chinese Prac Med*. 2012;7(8):16-17. doi:10.14163/j.cnki.11-5547/r.2012.08.024

38. Liu X, Chen L, Jiang C. Influence of family visit of community nurses for the elderly with dementia family caregivers' burden of care and quality of life of the elderly with dementia. *Nursing Research of China*. 2017;31(33):4206-4209. doi:10.3969/j.issn.1009-6493.2017.33.009

39. Lök N, Bademli K. Pilot testing of the "First You Should Get Stronger" program among caregivers of older adults with dementia. *Archives of Gerontology and Geriatrics*. 2017;68:84-89. doi:10.1016/j.archger.2016.09.006

40. Mahdavi B, Fallahi-Khoshknab M, Mohammadi F, Hosseini MA, Haghi M. Effects of Spiritual Group Therapy on Caregiver Strain in Home Caregivers of the Elderly with Alzheimer's Disease. *Archives of Psychiatric Nursing*. 2017;31(3):269-273. doi:10/gbhhs3

41. Norouzi M, Golzari M, Sohrabi F. Effectiveness of mindfulness based cognitive therapy on the quality of life, depression and burden of demented women caregivers. *Zahedan Journal of Research in Medical Sciences*. 2014;16(9):5-11. doi:10.13140/rg.2.1.2007.0882

42. Pahlavanzadeh S, Heidari FG, Maghsudi J, Ghazavi Z, Samandari S. The effects of family education program on the caregiver burden of families of elderly with dementia disorders. *Iranian journal of nursing and midwifery research*. 2010;15(3):102. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3093163/

43. Shata ZN, Amin MR, El-Kady HM, Abu-Nazel MW. Efficacy of a multi-component psychosocial intervention program for caregivers of persons living with neurocognitive disorders, Alexandria, Egypt: A randomized controlled trial. *Avicenna Journal of Medicine*. 2017;7(2):10. doi:10.4103/2231-0770.203610

44. Söylemez BA, Küçükgüçlü Ö, Buckwalter KC. Application of the Progressively Lowered Stress Threshold Model with Community-Based Caregivers: A

Randomized Controlled Trial. *J Gerontol Nurs*. 2016;42(7):44-54. doi:10.3928/00989134-20160406-01

45. He L, He R, Deng W, Zhang S. Effect of 3R therapy on cognitive function and human plasma amyloid protein β1-42 in patients with Alzheimer's disease. *Chinese Journal of Geriatric Heart Brain*. 2012;14(3):236-238. doi:10.3969/j.issn.1009-0126.2012.03.005

46. Sun X, Su H, Li X. Influence of nursing intervention on quality of life of patients with Alzheimer disease. *Chinese Nursing Research*. 2010;24(13):1167-1169. http://www.cqvip.com/qk/92301a/201005/33854421.html

47. Tan X, Lin C, Zhao L, et al. Effects of Early Family Nursing Intervention on Quality of Life of Patients with Alzheimer's Disease. *Nursing Journal of Chinese People's Liberation Army*. 2010;27(2A):176-179. doi:10.3969/j.issn.1008-9993.2010.03.005

48. Wang Z, Mu J bing, Liu R, Wang H xia. Community nursing intervention on quality of life in patients with senile dementia. *Modern Preventive Medicine*. 2010;37(5):840-844. https://www.cnki.com.cn/Article/CJFDTotal-XDYF201005016.htm

49. Wang LQ, Chien WT, Lee IYM. An experimental study on the effectiveness of a mutual support group for family caregivers of a relative with dementia in mainland China. *Contemporary Nurse*. 2012;40(2):210-224. doi:10.5172/conu.2012.40.2.210

50. Wang X, Zhang R, Liu W, et al. Effect of home care intervention on the rehabilitation outcome of patients with Alzheimer's disease. *Medical Journal of National Defending Forces in Southwest China*. 2014;24(01):85-86. doi:10.3969/j.issn.1004-0188.2014.01.036

51. Yang D liang, Li X, Su N, Ji W dong, Chen X xuan. Family medical intervention model of senile dementia with behavioral and psychological symptoms. *Journal of Shanghai Jiao Tong University (Medical Science)*. 2017;37(3):398-402. doi:10.3969/j.issn.1674-8115.2017.03.024

52. Zhao L, Tan X, Chen S. Observations on the effects of implementing home care interventions for patients with Alzheimer's disease. *Journal of Qiqihar Medical College*. 2010;31(6):948. http://www.cqvip.com/qk/98584x/201006/33936654.html

53. Bell ML, Fiero M, Horton NJ, Hsu CH. Handling missing data in RCTs; a review of the top medical journals. *BMC Medical Research Methodology*. 2014;14(1):118. doi:10.1186/1471-2288-14-118

54. Zhou N, Manser P. Does including machine learning predictions in ALS clinical trial analysis improve statistical power? *Annals of Clinical and Translational Neurology*. 2020;7(10):1756-1765. doi:10.1002/acn3.51140

APPENDIX 1 SEARCH STRATEGY

Appendix 1.1 Summary

	Condition	Ovid Medline	Embase	Global Health	APA PsycINFO	EBSCO CINAHL	Total
1	Include dementia-related terms	427238	723805	34331	185529	159287	
2	Include therapy-related terms	13930216	18103622	2046951	2371714	3652534	
3	Include RCT-related terms	5552799	5229127	632419	755859	982366	
4	Include LMIC-related terms	4046862	4706713	1683278	504266	705117	
5	Satisfy Conditions 1, 2, 3 & 4	7917	7967	1911	2359	1917	
6	Nonhuman experiments		31095663				
7	Human experiments		24104793				
8	Condition 6 results excluding Condition 7 results	5049406	6992081		379966	210517	
9	Condition 5 results excluding Condition 8 results	6646	7556		2301	1819	
10	Publication type is Editorial or Letter	2097191	1977082				
11	Condition 9 results excluding Condition 10 results	6594	7540				
12	Year of publication limited to 2019-2020	1156	1247	382	404	448	
	Total results:	1156	1247	382	404	448	
	Duplicates:	(0)	(357)	(143)	(212)	(272)	(984)
	Results for screening:	1156	890	239	192	176	2653

Searches updated and ran by Andra Fry on 21/09/2022.

Appendix 1.2 Ovid MEDLINE(R)

Ovid MEDLINE(R) ALL <1946 to September 20, 2022>		
Condition		Hit
1	exp Dementia/ or Pick Disease of the Brain/ or Huntington Disease/ or Lewy Body Disease/ or Cognitive Dysfunction/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*)).ti,ab,kf,kw.	427238
2	exp Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Antipsychotic Agents/ or exp Serotonin Uptake Inhibitors/ or exp Benzodiazepines/ or exp "Hypnotics and Sedatives"/ or exp Exercise/ or exp Exercise Therapy/ or exp Psychotherapy/ or exp Counseling/ or Psychosocial Support Systems/ or exp Complementary Therapies/ or Phototherapy/ or exp Advance Care Planning/ or Case Management/ or Caregivers/ed, px or Self-Help Groups/ or exp Social Support/ or Diagnosis, Computer-Assisted/ or Telemedicine/ or exp Computers, Handheld/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or	13930216

	(herb* adj3 (tea* or remedy or remedies or medicine*)) or gingko or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab,kf,kw.	
3	(Randomized Controlled Trial or Controlled Clinical Trial).pt. or (randomi#ed or placebo or randomly or trial or groups).ti,ab. or drug therapy.fs.	5552799
4	Developing Countries.sh,kf,kw. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,kf,kw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmics or third world or lami countr* or transitional countr*).ti,ab.	4046862
5	1 and 2 and 3 and 4	7917
6	exp Animals/ not Humans/	5049406
7	5 not 6	6646
8	(Comment or Editorial or Letter).pt.	2097191
9	7 not 8	6594
10	limit 9 to yr="2019 - 2020"	1156

Appendix 1.3 Embase

Embase <1974 to 2022 September 20>		
Condition		Hit
1	exp Dementia/ or Cognitive defect/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or	723805

	corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*)).ti,ab,kf,kw.	
2	exp Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Neuroleptic Agent/ or exp Serotonin Uptake Inhibitor/ or exp Benzodiazepine Derivative/ or Sedative Agent/ or exp Exercise/ or exp Kinesiotherapy/ or exp Psychotherapy/ or exp Counseling/ or Psychosocial Care/ or exp Alternative Medicine/ or Phototherapy/ or Advance Care Planning/ or Living Will/ or Case Management/ or Caregiver Support/ or Self Help/ or exp Social Support/ or Computer Assisted Diagnosis/ or Telemedicine/ or Telemonitoring/ or Telerehabilitation/ or Telediagnosis/ or Personal Digital Assistant/ or Smartphone/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or gingko or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab,kf,kw.	18103622
3	(Randomized Controlled Trial/ or Controlled Clinical Trial/ or random*.ti,ab. or Randomization/ or Intermethod Comparison/ or placebo.ti,ab. or (compare or compared or comparison).ti. or ((evaluated or evaluate or evaluating or assessed or assess) and (compare or compared or comparing or comparison)).ab. or (open adj label).ti,ab. or ((double or single or doubly or singly) adj (blind or blinded or blindly)).ti,ab. or Double Blind Procedure/ or parallel group*1.ti,ab. or (crossover or cross over).ti,ab. or ((assign* or match or matched or allocation) adj5 (alternate or group*1 or intervention*1 or patient*1 or subject*1 or participant*1)).ti,ab. or (assigned or allocated).ti,ab. or (controlled adj7 (study or design or trial)).ti,ab. or (volunteer or volunteers).ti,ab. or Human Experiment/ or trial.ti.) not (((random* adj sampl* adj7 (cross section* or questionnaire*1 or survey* or database*1)).ti,ab. not (Comparative Study/ or Controlled Study/ or randomi?ed controlled.ti,ab. or randomly assigned.ti,ab.)) or (Cross-Sectional Study/ not (Randomized Controlled Trial/ or Controlled Clinical Study/ or Controlled Study/ or randomi?ed controlled.ti,ab. or control group*1.ti,ab.)) or (((case adj control*) and random*) not randomi?ed controlled).ti,ab. or (systematic review not (trial or study)).ti. or (nonrandom* not random*).ti,ab. or random field*.ti,ab. or (random cluster adj3 sampl*).ti,ab. or ((review.ab. and review.pt.) not trial.ti.) or (we searched.ab. and (review.ti. or review.pt.)) or update review.ab. or (databases adj4 searched).ab. or ((rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset*1).ti. and Animal Experiment/) or (Animal Experiment/ not (Human Experiment/ or Human/)))	5229127
4	Developing Countries.sh,kf,kw. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or	4706713

	Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,kf,kw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmics or third world or lami countr* or transitional countr*).ti,ab.	
5	1 and 2 and 3 and 4	7967
6	exp Animal/ or exp Animal Experiment/ or Nonhuman/	31095663
7	exp Human/ or exp Human Experiment/	24104793
8	6 not 7	6992081
9	5 not 8	7556
10	(Editorial or Letter).pt.	1977082
11	9 not 10	7540
12	limit 11 to yr="2019 - 2020"	1247

Appendix 1.4 Global Health

Global Health <1973 to 2022 Week 37>		
Condition		Hit
1	exp Dementia/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*)).ti,ab.	34331
2	exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Neuroleptics/ or exp Selective Serotonin Reuptake Inhibitors/ or exp Benzodiazepines/ or exp Sedatives/ or Exercise/ or Physical Therapy/ or exp Psychotherapy/ or exp Counselling/ or "Complementary and Alternative Medicine"/ or Phototherapy/ or Self Help/ or Self Care/ or exp E-health/ or Mobile Telephones/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or gingko or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab.	2046951
3	Randomized Controlled Trials/ or (randomi#ed or placebo or randomly or trial or groups).ti,ab.	632419
4	(Low Income Countries or Lower-Middle Income Countries or Least Developed Countries).sh. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or	1683278

	Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmics or third world or lami countr* or transitional countr*).ti,ab.	
5	1 and 2 and 3 and 4	1911
6	limit 5 to yr="2019 - 2020"	382

Appendix 1.5 APA PsycInfo

APA PsycInfo <1806 to September Week 1 2022>		
Condition		Hit
1	exp Dementia/ or Huntington Disease/ or Cognitive Impairment/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*) or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr* adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild adj2 cognit* impair*)).ti,ab.	185529
2	exp Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Tranquilizing Drugs/ or exp Serotonin Reuptake Inhibitors/ or exp Benzodiazepines/ or exp Sedatives/ or exp Exercise/ or Movement Therapy/ or exp Psychotherapy/ or exp Counseling/ or Social Support/ or exp Alternative Medicine/ or Creative Arts Therapy/ or Mind Body Therapy/ or Phototherapy/ or Advance Directives/ or exp Case Management/ or Support Groups/ or exp Self-Help Techniques/ or exp Social Support/ or Computer Assisted Diagnosis/ or Computer Assisted Therapy/ or Telemedicine/ or exp Mobile Devices/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or	2371714

	traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or gingko or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab.	
3	exp Clinical Trials/ or Treatment Effectiveness Evaluation/ or (randomi#ed or placebo or randomly or trial or groups).ti,ab.	755859
4	Developing Countries.sh. or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,ti,ab,lo. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or lmic or lmics or third world or lami countr* or transitional countr*).ti,ab.	504266
5	1 and 2 and 3 and 4	2359
6	(animal not human).po.	379966
7	5 not 6	2301
8	limit 7 to yr="2019 - 2020"	404

Appendix 1.6 CINAHL EBSCO

CINAHL EBSCO		
Condition		Hit
S1	(MH "Dementia+") or (MH "Huntington's Disease") or (MH "Cognition Disorders") or TI ((dement* or amentia* or (major N2 cognit* N2 disorder*) or alzheimer* or alzeimer* or (cortical N3 sclerosis) or ((encephalopath* or cogniti* or neurocogniti*) N3 (aids or "acquired immun#deficiency syndrome*" or "acquired immun# deficiency syndrome*" or hiv or "human immun#deficiency virus*" or "human immun# deficiency virus*")) or "pick* disease*" or (lobar N2 atroph* N2 brain) or "Huntington* disease*" or "Huntington* chorea" or ("Lewy bod*" N2 disease*) or (cerebr* N2 deteriorat*) or (cerebr* N2 insufficien*) or ((frontotemporal or "fronto temporal" or corticobasal or "cortico basal" or "frontal lobe*") N3 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) N2 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild N1 "cognit* impair*"))) OR AB ((dement* or amentia* or (major N2 cognit* N2 disorder*) or alzheimer* or alzeimer* or (cortical N3 sclerosis) or	159287

	((encephalopath* or cogniti* or neurocogniti*) N3 (aids or "acquired immun#deficiency syndrome*" or "acquired immun# deficiency syndrome*" or hiv or "human immun#deficiency virus*" or "human immun# deficiency virus*")) or "pick* disease*" or (lobar N2 atroph* N2 brain) or "Huntington* disease*" or "Huntington* chorea" or ("Lewy bod*" N2 disease*) or (cerebr* N2 deteriorat*) or (cerebr* N2 insufficien*) or ((frontotemporal or "fronto temporal" or corticobasal or "cortico basal" or "frontal lobe*") N3 (degenerati* or dysfunction*)) or ((cognit* or memory or cerebr*) N2 (declin* or impair* or los* or deteriorat* or degenerat* or insufficen*)) or MCI or (mild N1 "cognit* impair*")))	
S2	(MH "Cognitive Therapy+") or (MH "Drug Therapy+") or (MH "Cholinesterase Inhibitors+") or (MH "Antipsychotic Agents+") or (MH "Serotonin Uptake Inhibitors+") or (MH "Antianxiety Agents, Benzodiazepine+") or (MH "Hypnotics and Sedatives+") or (MH "Exercise+") or (MH "Therapeutic Exercise+") or (MH "Psychotherapy+") or (MH "Counseling+") or (MH "Support, Psychosocial+") or (MH "Alternative Therapies+") or (MH "Advance Care Planning") or (MH "Advance Directives+") or (MH "Case Management") or (MH "Patient Care Plans+") or (MH "Caregivers/ED/PF") or (MH "Support Groups") or (MH "Diagnosis, Computer Assisted") or (MH "Telehealth+") or (MH "Computers, Hand-Held+") or TI ((intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* N2 (therap* or training or rehab*)) or "cognit* psycho therap*" or "cognit* psychotherap*" or drug* or medicine* or pharmacotherap* or "pharmaco* therap*" or "cholinesterase agent*" or "cholinesterase inhibitor*" or (tranquili* N2 (agent* or drug*)) or antipsychotic* or (neuroleptic N2 (agent* or drug*)) or "serotonin uptake inhibitor*" or "serotonin reuptake inhibitor*" or ssri* or benzodiazepine* or (sedative N2 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or "physical activit*" or "physical training" or aerobic* or arobic* or kinesi#therap* or "kinesi# therap*" or (social N2 (activit* or engag* or stimul*)) or "psycholog* therap*" or psychotherap* or ((behavio#r* or conditioning) N2 therap*) or counsel#ing or ((psychosocial or "psycho social") N2 (support or interven* or care)) or ((alternative or compl#ment* or traditional) N2 (medicine* or therap*)) or acupunct* or (herb* N2 (tea* or remedy or remedies or medicine*)) or gingko or homeopath* or ((music or art or aroma or light or photo or pet or pets) N2 therap*) or aromatherap* or phototherap* or massage* or (mind N2 body) or (advance# N2 (care or medical or healthcare) N2 plan*) or (decision* N2 (aid* or support)) or (communicati* N2 skill* N2 training) or ("dementia care" N2 map*) or ((person* or patient*) N2 cent* N2 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) N2 educat*) or telemedicine or "tele medicine" or (smart N2 (phone* or device* or tablet*)) or smartphone* or "cognitive aid*" or reminder* or robot*)) OR AB ((intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* N2 (therap* or training or rehab*)) or "cognit* psycho therap*" or "cognit* psychotherap*" or drug* or medicine* or pharmacotherap* or "pharmaco* therap*" or "cholinesterase agent*" or "cholinesterase inhibitor*" or (tranquili* N2 (agent* or drug*)) or antipsychotic* or (neuroleptic N2 (agent* or drug*)) or "serotonin uptake inhibitor*" or "serotonin reuptake inhibitor*" or ssri* or benzodiazepine* or (sedative N2 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or "physical activit*" or "physical training" or aerobic* or arobic* or kinesi#therap* or "kinesi# therap*" or (social N2 (activit* or engag* or stimul*)) or "psycholog* therap*" or psychotherap* or ((behavio#r* or conditioning) N2 therap*) or counsel#ing or ((psychosocial or "psycho social") N2 (support or interven* or care)) or ((alternative or compl#ment* or traditional) N2 (medicine* or therap*)) or acupunct* or (herb* N2 (tea* or remedy or remedies or medicine*)) or gingko or homeopath* or ((music or art or aroma or light or photo or pet or pets) N2 therap*) or aromatherap* or phototherap* or massage* or (mind N2 body) or (advance# N2 (care or medical or healthcare) N2 plan*) or (decision* N2 (aid* or support)) or (communicati* N2 skill* N2 training) or ("dementia care" N2 map*) or ((person* or patient*) N2 cent* N2 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) N2 educat*) or telemedicine or "tele medicine" or (smart N2 (phone* or device* or tablet*)) or smartphone* or "cognitive aid*" or reminder* or robot*))	3652534
S3	(MH "Randomized Controlled Trials") or (MH "Double-Blind Studies") or (MH "Single-Blind Studies") or (MH "Random Assignment") or (MH "Pretest-Posttest Design+") or (MH "Cluster Sample+") or TI (randomised or randomized) or AB random* or TI trial or (MH "Sample Size" and AB (assigned OR allocated OR control)) or (MH "Placebos") or (MH "Crossover Design") or (MH "Comparative Studies") or PT "randomized controlled trial" or AB ((control W5 group) or (cluster W3 RCT))	982366
S4	(MH "Developing Countries")	19971
S5	TI (Africa* or Asia* or Caribbean* or "West Indies" or "South America*" or "Latin America*" or "Central America*" or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or "Burkina Faso" or "Burkina Fasso" or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or "Cape Verd*" or "Cabo Verde" or "Central African Republic" or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or "Comoro Islands" or Comores or Comoran or Mayotte or Congo* or "Costa Rica*" or "Cote d'Ivoire" or "Ivory Coast" or Ivorian* or "Cook Islands" or Cuba* or Croat* or Djibouti* or Dominica* or "East Timor" or "East Timur" or "Timor Leste" or Timorese or Ecuador* or Equador* or Egypt* or "El Salvador" or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or "Georgia Republic" or Georgian or Abkhazia* or Abchasia* or "South	679546

	Ossetia*" or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or "Guinea Bissau" or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or "Lao PDR" or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or "Marshall Islands" or Marshallese or Mauritania* or Mauriti* or "Agalega Islands" or Mexico or Mexican or Micronesia* or "Middle East*" or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or "Papua New Guinea*" or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or "Saint Helen*" or "St Helen*" or "Saint Kitts" or "St Kitts" or Kittian or Nevis* or "Saint Lucia*" or "St Lucia*" or "Saint Vincent" or "St Vincent" or Vicentian or Grenadines or Samoa* or "Sao Tome*" or Senegal* or Serbia* or Seychell* or "Sierra Leone*" or "Sri Lanka*" or Ceylon or "Solomon Island*" or Somali* or "South Africa*" or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or "Viet Nam" or "Wallis Futuna" or "West Bank" or Yemen* or Zambia* or Zimbabwe*) OR AB (Africa* or Asia* or Caribbean* or "West Indies" or "South America*" or "Latin America*" or "Central America*" or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or "Burkina Faso" or "Burkina Fasso" or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or "Cape Verd*" or "Cabo Verde" or "Central African Republic" or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or "Comoro Islands" or Comores or Comoran or Mayotte or Congo* or "Costa Rica*" or "Cote d'Ivoire" or "Ivory Coast" or Ivorian* or "Cook Islands" or Cuba* or Croat* or Djibouti* or Dominica* or "East Timor" or "East Timur" or "Timor Leste" or Timorese or Ecuador* or Equador* or Egypt* or "El Salvador" or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or "Georgia Republic" or Georgian or Abkhazia* or Abchasia* or "South Ossetia*" or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or "Guinea Bissau" or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or "Lao PDR" or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or "Marshall Islands" or Marshallese or Mauritania* or Mauriti* or "Agalega Islands" or Mexico or Mexican or Micronesia* or "Middle East*" or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or "Papua New Guinea*" or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or "Saint Helen*" or "St Helen*" or "Saint Kitts" or "St Kitts" or Kittian or Nevis* or "Saint Lucia*" or "St Lucia*" or "Saint Vincent" or "St Vincent" or Vicentian or Grenadines or Samoa* or "Sao Tome*" or Senegal* or Serbia* or Seychell* or "Sierra Leone*" or "Sri Lanka*" or Ceylon or "Solomon Island*" or Somali* or "South Africa*" or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or "Viet Nam" or "Wallis Futuna" or "West Bank" or Yemen* or Zambia* or Zimbabwe*) OR MH (Africa* or Asia* or Caribbean* or "West Indies" or "South America*" or "Latin America*" or "Central America*" or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or "Burkina Faso" or "Burkina Fasso" or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or "Cape Verd*" or "Cabo Verde" or "Central African Republic" or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or "Comoro Islands" or Comores or Comoran or Mayotte or Congo* or "Costa Rica*" or "Cote d'Ivoire" or "Ivory Coast" or Ivorian* or "Cook Islands" or Cuba* or Croat* or Djibouti* or Dominica* or "East Timor" or "East Timur" or "Timor Leste" or Timorese or Ecuador* or Equador* or Egypt* or "El Salvador" or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or "Georgia Republic" or Georgian or Abkhazia* or Abchasia* or "South Ossetia*" or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or "Guinea Bissau" or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or "Lao PDR" or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or	

	Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or "Marshall Islands" or Marshallese or Mauritania* or Mauriti* or "Agalega Islands" or Mexico or Mexican or Micronesia* or "Middle East*" or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegr* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or "Papua New Guinea*" or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or "Saint Helen*" or "St Helen*" or "Saint Kitts" or "St Kitts" or Kittian or Nevis* or "Saint Lucia*" or "St Lucia*" or "Saint Vincent" or "St Vincent" or Vicentian or Grenadines or Samoa* or "Sao Tome*" or Senegal* or Serbia* or Seychell* or "Sierra Leone*" or "Sri Lanka*" or Ceylon or "Solomon Island*" or Somali* or "South Africa*" or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadzhikistan or Tadjikistan or Tajik or Tadzhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or "Viet Nam" or "Wallis Futuna" or "West Bank" or Yemen* or Zambia* or Zimbabwe*)	
S6	TI ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income" or underserved or "under served" or deprived or poor*) N1 (countr* or nation# or population# or world)) or ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income") N1 (economy or economies)) or (low* N1 (gdp or gnp or "gross domestic" or "gross national")) or (low N2 middle N2 countr*) or lmic or lmics or "third world" or "lami countr*" or "transitional countr*") OR AB ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income" or underserved or "under served" or deprived or poor*) N1 (countr* or nation# or population# or world)) or ((developing or "less* developed" or "under developed" or underdeveloped or "middle income" or "low* income") N1 (economy or economies)) or (low* N1 (gdp or gnp or "gross domestic" or "gross national")) or (low N2 middle N2 countr*) or lmic or lmics or "third world" or "lami countr*" or "transitional countr*")	46191
S7	S4 or S5 or S6	705117
S8	S1 AND S2 AND S3 AND S7	1917
S9	(MH "Animals+" OR MH "Animal Studies" OR TI "animal model*") NOT MH "Human"	210517
S10	S8 NOT S9	1819
S11	Limiters - Published Date: 20190101-20211231	448

APPENDIX 2 SELECTED CHARACTERISTICS OF INCLUDED STUDIES

Appendix 2.1 Studies from the 2022 search

Study identifier	Country	Descriptive name of the intervention	Intervention approach(es)	Intervention frequency	Randomisation unit	Sample size (Baseline)		Intervention duration	Follow-up duration (from baseline)
						Control	Intervention		
Chen 2020[30]	China	Extended Nursing	General support	N/A	Individual	51	51	8 weeks	1 year
Govindakumari 2020[31]	Iran	Resilience Education	Cognitive-behavioural intervention	Weekly	Individual	27	27	8 weeks	8 weeks
Ghaffari 2019[32]	India	Home-based cognitive training program	Cognitive and functional training	Twice a week	Individual	157	157	1 month	1 month
Hinton 2020[33]	Vietnam	Multi-component interventions based on REACH	Multi-component intervention (PEI+CBT+CCM+CFT)	Weekly	Community	26	25	3 months	3 months
Pan 2019[34]	China	Nurse-led cognitive behavioural sessions and subsequent consultation calls	Cognitive-behavioural intervention	Monthly	Individual	45	54	5 months	7 months
Uyar 2019[14]	Turkey	Multicomponent psychosocial intervention	Multi-component intervention (PEI+GS)	N/A	Individual	26[b]	28[b]	16 weeks	16 weeks
Zarepour 2020[35]	Iran	On-site educational intervention along with informational support on the anxiety of family	Psychoeducational intervention	7 sessions in 4 weeks	Individual	35	35	4 weeks	4weeks + 1 month 4weeks + 3 months

Appendix 2.2 Studies from the 2019 meta-analysis

Study identifier	Country	Descriptive name of the intervention	Intervention approach(es)	Intervention frequency	Randomisation unit	Sample size (Baseline)		Intervention duration	Follow-up duration (from baseline)
						Control	Intervention		
Amit 2008[36]	India	Home Care Program for supporting caregivers of PwD	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	26	33	6 months	3 months 6 months

[b] The authors stated in the Abstract that “the intervention (n=31) and control groups (n=30)” but in the main text, the authors stated that “the intervention (n=30) and control groups (n=31)”. Thus, we consider this to be a typo.

Guerra 2011[15]	Peru	'Helping Carer to Care' training	Multi-component intervention (PEI+CBT)	Weekly	Individual	29	29	5 weeks	6 months
JIANG 2012[37]	China	Whole course rehabilitation nursing intervention	Multi-component intervention (PEI+CCM)	N/A	Individual	48	46	1 year	1 year
Liu 2017[38] [c]	China	Family visits of community nurses	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	39	39	3 months	3 months
Lök 2017[39]	Turkey	"First You Should Get Stronger" Caregiving Program	Psychoeducational intervention	Weekly	Individual	20	20	7 weeks	7 weeks
Mahdavi 2017[40]	Iran	Spiritual Group Therapy	Miscellaneous	Weekly	Individual	31	28[d] / 30[e]	5 weeks	5 weeks
Mercedeh 2014[41]	Iran	Mindfulness based cognitive therapy	Cognitive-behavioural intervention	Weekly	Individual[f]	10	10	8 weeks	8 weeks 8 weeks + 2 months
Orawan 2018[11]	Thailand	Enhancing Positive Aspects of Caregiving Program	Cognitive-behavioural intervention	Weekly	Individual	36	36	7 weeks	8 weeks 12 weeks 20 weeks
Pahlavanzadeh 2010[42]	Iran	Family education program	Psychoeducational intervention	Weekly	Individual	25	25	5 weeks	5 weeks 5 weeks + 1 month
Salamizadeh 2017[12]	Iran	Spiritual Care Education	Miscellaneous	Weekly	Individual [f]	28	26	5 weeks	5 weeks
Shata 2017[43]	Egypt	A multi-component psychosocial intervention program	Multi-component intervention (PEI+CBT+GS)	Weekly	Individual	59	55	8 weeks	8 weeks 8 weeks + 3 months
Söylemez 2016[44]	Turkey	Progressively Lowered Stress Threshold-based intervention	Multi-component intervention (PEI+CCM+GS)	Bi-weekly	Individual	34	35	3 months	5 months + 2 weeks[g]
SU 2012[45]	China	Comprehensive nursing intervention	Multi-component intervention (PEI+CCM)	N/A	Individual	20	20	14 days	14 days + 6 months
SUN 2010[46]	China	Nursing intervention	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	19	19	1 year	1 year
TAN 2010[47]	China	Early family intervention	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	26	26	1 year	1 year

[c] The study identifier in the 2019 meta-analysis records was "Lin 2017", which proved to be a typo during full-text screening.
[d] Control group 1: Group sessions without spiritual therapy
[e] Intervention group: Spiritual group therapy
[f] The authors claimed that the study was a quasi-experiment, but the study randomised its participants.
[g] The second home visits took place two weeks after the first visit. The third home visit took place 2 months after the second visit. The measurement occurred 3 months after the third visit.

Wang 2010b[48]	China	Community nursing intervention	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	40	40	1 year	1 year
Wang 2012e[49]	China	Mutual support group	Multi-component intervention (CCM+GS)	Bi-weekly	Individual	39	39	24 weeks	24 weeks 24 weeks + 1 month
Wang 2014a[50]	China	Home care intervention	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	61	62	1 year	6 months 12 months
Yang 2017a[51]	China	Family medical intervention model	Multi-component intervention (PEI+CCM+GS)	N/A	Community	70	71	1 year	6 months 1 year
ZHAO 2010[52]	China	Family nursing intervention	Multi-component intervention (PEI+CCM+GS)	N/A	Individual	25	25	1 year	1 year

APPENDIX 3 RISK OF BIAS INFORMATION

Appendix 3.1 Chen 2020

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: Low risk **Allocation sequence concealment? Y** **Randomisation? Y** The authors used the random number method to generate the allocation sequence, which is a centralised concealed procedure. **Baseline differences? N** According to the authors, "In the control group, there were 23 males and 28 females (aged 63-82 y, with a mean age of 71.43±5.14 y), the course of disease was from 4 to 12 y (mean duration of 7.44±2.15 y). In the observation group, there were 24 males and 27 females (aged 62-84 y, with a mean age of 72.38±4.27 y); the course of disease was from 5 to 10 y (mean duration of 7.13±1.64 y). There was no statistical difference in the baseline data between these two groups."	**Conclusion:** Some concerns **Participants aware? Y** The "families voluntarily participated in the study and signed an informed consent." Thus, the participants were aware of the intervention. **Personnel aware? Y** According to the authors, the control group only received routine care, while the interventionists "participated in the study actively and were trained with the extended nursing scheme." Thus, they should know that they were in the intervention group. **Deviation from intervention due to trial context? PN** The authors mentioned that "Enhancement of communication among doctors, nurses and patients and if the patients or nursing staff do not appear in group chat for a month, the nursing team members inquired the reasons, informed the significance of active participation in group chat and recommended participation." Thus, there could be deviations during the study. However, this could happen if there was not any trial in place. **Appropriate analysis to estimate the effects? Y** The study used ITT analysis and reported no subjects with missing data. Thus, the method was appropriate.	**Conclusion:** Low risk **Outcome data for all participants? Y** The authors did not report any missing data or describe any dropout. The analysed sample sizes in Table 1, Table 2 and Table 3 were all consistent with the initial sample size (n = 51), which also suggests no missing data.	**Conclusion:** High risk **Inappropriate outcome measurement? N** **Measurement differed between groups? N** "The behavioral pathology assessment scale of Alzheimer's disease (BEHAVE-AD) [8] was used in this study. The behavioral and psychological symptoms of the patients were scored by the doctors. […]Montreal Cognitive Assessment Scale (MoCA)[9] and mini mental state examination scale (MMSE) were used to evaluate the cognitive function ***of the 2 groups***." All the scales used are well-established methods of outcome measurement. Both groups used the same scales. **Assessors aware? NI** The authors did not mention who performed the measurement and whether they were masked. **Assessment have been influenced by knowledge of intervention? NI** **Assessment likely influenced by knowledge of intervention? NI** The authors did not mention who performed the measurement and whether they were masked. There is insufficient information.	**Conclusion:** Some concerns **Selected from multiple eligible outcomes? N** MoCA and MMSE were both measurement of care recipients' cognitive function used in this study. They were all reported. **Selected from multiple eligible analyses? NI** We didn't find any pre-specified protocol or analysis plan for the study. Analysis intentions were not described in this study. Thus, there is not enough information to enable judgement. **Consistent with pre-specified analysis plan? NI** However, we were able to acquire the information of the fund used in this study, yet the fund was not intended for this topic of this study and the study was not reported to the funder.	High risk

Appendix 3.2 Ghaffari 2019

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: High risk **Allocation sequence concealment? NI** **Randomisation? NI** The eligible participants were "grouped into intervention and control groups by a block-randomized allocation method", yet more details may be required. There is no clear mentioning of a central computerised programme for randomisation, so the randomisation may not be consistent with the best practice. Also, it is not clear whether the randomisation was concealed. **Baseline differences? Y** The authors used block randomisation with additional adjustment for age, sex, marital status, occupation, education level, income level, caring duration, smoking situation, and family relationship. However, it was unclear how the comparison in Table 2 was made. When using chi-square and Fisher's exact tests to recalculate the result from Table 2, I am not able to get the same result.	**Conclusion:** High risk **Participants aware? Y** **Personnel aware? Y** The participants were aware of the assigned intervention, as they had written consent of the experiment and the control group had no intervention at all. Same case for the people delivering the intervention. **Deviation from intervention due to trial context? NI** The authors did not mention any deviation from planned interventions or any measures to avoid or correct for the deviation. **Deviations balanced between groups? NI** The authors didn't further describe the deviation with relevant data. Thus, there is insufficient information. **Appropriate analysis to estimate the effects? NI** The authors used ITT analysis, but it is unclear how the authors dealt with participants with missing data.	**Conclusion:** High risk **Outcome data for all participants? N** "In total, 64 family caregivers of older adults with Alzheimer's disease completed an initial eligibility assessment. However, 10 family caregivers were excluded from the study based on the exclusion criteria (n = 4) or due to declining to participate (n = 6). […] Two participants in the intervention group missed two or more program sessions and two participants in the control group missed assessments due to the death of their patients during the program. Therefore, of 54 family caregivers, 50 (92.6%) completed all follow-up assessments (25 in the intervention group and 25 in the control group)." **Evidence that result is not biased? N** The authors did not mention such evidence. **Missingness could depend on true value? NI** The authors didn't further explain the exclusion criteria and why people declined to participate. **Likely that missingness depended on true value? PY** The reported reasons for missing data differed in control and intervention groups. "Two participants in the intervention group missed two or more program sessions and two participants in the control group missed assessments due to the death of their patients during the program."	**Conclusion:** High risk **Inappropriate outcome measurement? N** Connor-Davidson Resilience Scale and General Health Questionnaire (GHQ) are well-established metrics for such studies. **Measurement differed between groups? N** Both control and intervention groups used the same questionnaires. **Assessors aware? PY** **Assessment have been influenced by knowledge of intervention? PY** The self-assessments may be compromised as the patient knew that they were in the intervention group and thus had a positive psychological suggestion **Assessment likely influenced by knowledge of intervention? Y** There is no direct evidence of such influence. However, according to the guideline of RoB 2 tool, "when there are strong levels of belief in either beneficial or harmful effects of the intervention, it is more likely that the outcome was influenced by knowledge of the intervention received." As the intervention group was believed to be likely effective, it was likely that the knowledge affected the assessment.	**Conclusion:** Some concerns **Selected from multiple eligible outcomes? NI** The study reported caregiver mental health but not caregiver resilience scores, while both were listed as outcomes in the study protocol avaiialble at https://www.irct.ir/trial/23558 However, the trial was updated retrospectively. Thus, it is unclear whether the study protocol was the same as the original protocol. **Selected from multiple eligible analyses? NI** The analysis intentions or plan were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened.	High risk

Appendix 3.3 Govindakumari 2020

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: Some concerns **Randomisation? NI** The authors claimed that the study was randomised, but no further information about randomisation was given. **Allocation concealment? NI** The authors claimed that the study was randomised, but no further information about allocation concealment was given. **Baseline differences: NI** Only outcomes were measured at baseline, whereas other relevant information, e.g. demographic information, was lacking.	**Conclusion:** High risk **Participants aware? NI** **Personnel aware? NI** The authors only mentioned that "cognitive training was given to all eligible subjects individually two sessions weekly continuously for a period of 1 month i.e. a total of 8 sessions. Each session will take approximately 1 hour." They did not mention any placebo or sham intervention or describe what they did to the control group. We are unclear whether the participants or interventionists were blinded from. **Deviation from intervention due to trial context? NI** The authors did not mention any deviation from planned interventions or any measures to avoid or correct for the deviation. **Deviations affect outcome? NI** Not mentioned in the paper. **Deviation balanced between groups? NI** Not mentioned in the paper. **Appropriate analysis to estimate the effects? NI** Unable to tell whether the study used ITT or PP analysis, due to a lack of information regarding deviations.	**Conclusion:** High risk **Outcome data for all participants? NI** The authors did not report any missing data or describe any dropout. **Evidence that result is not biased? N** The authors did not mention such evidence. **Missingness could depend on true value? NI** The authors did not describe missingness. Thus, the information is lacking. **Likely that missingness depended on true value? NI** The authors did not describe missingness. Thus, the information is lacking.	**Conclusion:** High risk **Inappropriate outcome measurement? NI** The authors did not describe the scales used to measure the outcomes. Thus, it is unclear whether the measurement was inappropriate? **Measurement differed between groups? NI** The authors did not provide how measurement was done. Thus, there is no sufficient information to enable judgement. **Assessors aware? NI** The authors did not describe how measurement was done. Thus, there is no sufficient information to enable judgement. **Assessment have been influenced by knowledge of intervention? NI** The authors did not describe how measurement was done. Thus, there is no sufficient information to enable judgement. **Assessment likely influenced by knowledge of intervention? NI** The authors did not describe how measurement was done. Thus, there is no sufficient information to enable judgement.	**Conclusion:** Some concerns **Selected from multiple eligible outcomes? NI** The authors did not describe how outcome measurement was done. Thus, there is no sufficient information to enable judgement. **Selected from multiple eligible analyses? PN** The authors used both paired and unpaired t-test and reported all analysis results. However, the authors did not test the normality assumption for t-test. Also, the caregiver burden surprisingly saw significant decline in the control group.	High risk

Appendix 3.4 Hinton 2020

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: Low risk **Allocation sequence concealment? Y** **Randomisation? Y** The authors described that "randomization was conducted by one of the study investigators residing in the United States through the flip of a coin." Thus, the allocation sequence was concealed and random. **Baseline differences: N** "There were no statistically significant differences in the characteristics at baseline except. for the number of hours providing care to the person with dementia each week, which was significantly higher in the enhanced control condition compared with the intervention group (P = .03)."	**Conclusion:** Some concerns **Participant aware? Y** "At the time of consent, participants were informed about their allocation to either the enhanced control or the intervention condition." **Personnel aware? Y** The study was open label, so the interventionists were not blinded from the intervention. **Deviation from intervention: Y** "Based on a review of standardized treatment delivery forms completed by the interventionists at the end of each session for the REACH VN groups, all required elements were completed for 98% of session 1, 96% for session 2, 92% for session 3, 87% for session 4, and 100% of final (closure) sessions." **Deviations affect outcome? N** There is no evidence to indicate so. As the attendance rate was at least 87%, with the non-attendance even less than attribution. Thus, the outcome may not be significantly affected. **Appropriate analysis to estimate the effects? Y** The study mentioned that "primary analysis of the intervention effect used an intent-to-treat approach, in which groups were compared based on randomization assignment." Of 60 participants, only 51 participants who completed the study were analysed, with those with missing data excluded. Thus, this was an appropriate analysis.	**Conclusion:** Some concerns **Outcome data for all participants? N** The authors described dropouts in the study. "Of the 60 participants randomized (across nine clusters), 51 completed the study (26 in the control group and 25 in REACH VN intervention)." **Evidence that result is not biased? N** The authors did not mention such evidence. **Missingness could depend on true value? Y** According to the authors, "reasons for caregiver attrition included care recipient passing away (n = 5, two in intervention and three in control), caregiver decided not to participate (n = 3, due to lack of time and other reasons), and one caregiver was found to be ineligible due to significant cognitive impairment." These reasons were related to care recipient health and could depend on its true value. **Likely that missingness depended on true value? N** Missingness was largely balanced between intervention groups and there was no direct evidence for dependency on true value.	**Conclusion:** Low risk **Inappropriate outcome measurement? N** ZBI, GHQ-28, AD knowledge questionnaire were all established outcome measurement methods. **Measurement differed between groups? N** The outcomes were measured by the same scales across groups. **Assessors aware? PN** "Baseline assessments were conducted by research staff who were not masked to allocation," yet "assessments at 3 months were conducted in the participants' homes by bachelor's level research staff masked to allocation". The baseline data were expected to be similar due to randomisation, which minimised the impact of non-blinding of assessors. Post-intervention measurement was blinded. Thus, the risk of bias for blinding of assessors was low.	**Conclusion:** Low risk **Selected from multiple eligible outcomes? N** The authors used ZBI, GHQ-28, AD knowledge questionnaire to measure different domains. Thus, there was not a domain measured in multiple ways. **Selected from multiple eligible analyses? N** The authors used only the adjusted model, which was a commune, mixed-effects models that included a cluster-specific random effect, to analyse the outcome, which was also the only way to measure the effects as stated in their study protocol. **Consistent with pre-specified analysis plan? Y** The study protocol is available at https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6211867/ The analysis plan is available at https://clinicaltrials.gov/ct2/show/NCT03587974 The plan was updated on 1 March 2020, which was later than the study completion date 31 October 2018. All outcomes were reported as stated in the methods section and the information on clinicaltrial.org.	Some concerns

Appendix 3.5 Pan 2019

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: Low risk **Allocation sequence concealment? Y** **"**The researcher was initially blinded to the study allocation but supervised the intervention; data collectors were blinded to the treatment assignments; and caregivers received a sealed letter with an allocated number and were informed of their treatment when the intervention began.**"** **Randomisation? Y** "After the initial assessment, caregivers were randomly allocated by a data processor into either the intervention or control group via a computerized random number, with 56 caregivers in each group." **Baseline differences? NI** Table 2 show the baseline data in the two groups, which doesn't show any significant differences.	**Conclusion:** High risk **Participants aware? Y** "Caregivers received a sealed letter with an allocated number and were informed of their treatment when the intervention began" **Personnel aware? Y** "The researcher was initially blinded to the study allocation but supervised the intervention." **Deviation from intervention due to trial context? PY** The authors mentioned that "although nurses were instructed to follow the protocol, minimal flexibility was allowed to adapt to individual caregivers. […] At the initial session, caregivers agreed to make up a session within a few days after a missed appointment." **Deviations affect outcome? NI** The authors didn't further describe the deviation with relevant data. **Deviations balanced between groups? NI** The authors didn't further describe the deviation with relevant data. **Appropriate analysis to estimate the effects? PN** The guideline of RoB2 tool suggests that "both intention-to-treat (ITT) analyses and modified intention-to-treat (mITT) analyses excluding participants with missing outcome data should be considered appropriate."[16(p2)] The study used ITT analysis, where missing data were imputed using the	**Conclusion:** High risk **Outcome data for all participants? N** "Family caregivers of PWD (N = 112; intervention group n = 56, control group n = 56) were recruited from the 276 caregivers referred and were evaluated at Time 1. At Time 2, 99 caregivers (intervention group n = 54, control group n = 45) completed the survey; and at Time 3, 82 caregivers (intervention group n = 47, control group n = 35) completed the full three evaluations." **Evidence that result is not biased? N** The authors did not mention such evidence. **Missingness could depend on true value? NI** "Among dropouts in the intervention group, two caregivers' PWD died, six caregivers discontinued caregiving, and one caregiver relocated; in the control group, seven caregivers' PWD died, three caregivers discontinued caregiving, eight caregivers relocated, and three caregivers refused the repeated surveys." However, the authors didn't describe why people gave up caregiving or refused repeated surveys. **Likely that missingness depended on true value? Y** There were much more dropouts in control, compared with the intervention group, suggesting a risk of bias related to the true value of the outcome.	**Conclusion:** High risk **Inappropriate outcome measurement? N** The authors used CES-D, 15-item Mutuality Scale, the Simplified Coping Scale, 14-item Activities of Daily Living (ADL) scale, which were all validated methods, with the Cronbach's alpha of its Chinese version provided. These show that these methods were appropriate. **Measurement differed between groups? N** Both control and intervention groups used the same questionnaires. **Assessors aware? PY** "Data collectors were blinded to the treatment assignments." However, self-report scales such as CES-D must be assessed by the participants themselves. **Assessment have been influenced by knowledge of intervention? Y** Yes, the caregivers may undergo placebo effect. **Assessment likely influenced by knowledge of intervention? PY** According to the guideline of RoB 2 tool, "when there are strong levels of belief in either beneficial or harmful effects of the intervention, it is more likely that the outcome was influenced by knowledge of the intervention received." As the intervention group was believed to be likely effective, it was likely that the knowledge affected the assessment.	**Conclusion:** Some concerns **Selected from multiple eligible outcomes? NI** The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. **Selected from multiple eligible analyses? NI** The analysis intentions were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened.	High risk

	missForest method. Thus, the analysis was different from the guideline. Moreover, it should be noted that there were more dropouts in the control group than in the intervention group. Thus, the algorithm didn't have a balanced impact on the data of two groups. **Substantial impact of inappropriate analysis? Y** The authors should have performed a sensitivity analysis to examine the robustness of the assumptions made in the primary analysis,[53] and to examine whether the machine learning algorithm improves statistical power in an anticipated way.[54]				

Appendix 3.6 Uyar 2019

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: High risk **Allocation sequence concealment? NI** The authors didn't mention whether the allocation was concealed, nor was there any evidence to indicate this. **Randomisation? Y** "Randomisation criteria were dementia type, dementia stage and caregivers' education level. We stratified the patients and caregivers based on the education status of caregivers, considering the randomisation criteria. Using ***simple random placement method***, we determined groups of caregivers to be assigned to the groups. The patient–caregiver pairs were assigned to the randomised IG (n = 30) and CG (n = 31)" **Baseline differences? Y** Table 2 shows the baseline demographic data. Marital status showed significant difference between the control and intervention groups.	**Conclusion:** High risk **Participants aware? PY** "We obtained written informed consent from all caregivers and some of the patients through the Volunteer Form." Thus, the participants knew the intervention, which was different from the routine care they typically received. **Personnel aware? PY** The interventionists provided structured training and support, which was different from the routine care they typically provided. Thus, they should be aware of the intervention. **Deviation from intervention due to trial context? NI** **Deviations affect outcome? NI** **Deviations balanced between groups? NI** The authors didn't further describe the deviation with relevant data. Thus, there is insufficient information. **Appropriate analysis to estimate the effects? Y** The study used ITT analysis, excluding the participants with missing data. This was an appropriate method according to the guideline of RoB 2 tool, which says, "both intention-to-treat (ITT) analyses and modified intention-to-treat (mITT) analyses excluding participants with missing outcome data should be considered appropriate."	**Conclusion:** Low risk **Outcome data for all participants? N** Figure 1 shows that the control group had two dropouts and two deaths, while the intervention group had 1 deaths and 2 who refused. **Evidence that result is not biased? N** The authors did not mention such evidence. **Missingness could depend on true value? N** There is no evidence to indicate this, as the attribution in two groups were similar.	**Conclusion:** High risk **Inappropriate outcome measurement? N** All the measurement methods were well-established with the local version's Cronbach's Alpha clearly stated. E.g. "Quality of Life Scale SF36 comprises eight sub- dimensions and defines two summary areas: mental health and physical health. The total score can be 0–100. Koçyiğit et al. (1999) assessed the Turkish reliability and validity of the scale. Cronbach's α of the scale was 0.73–0.76 for sub-dimensions of the scale (14); in this study, it was 0.73–0.88." **Measurement differed between groups? PY** The authors used different scales measuring quality of life in patients and caregivers. **Assessors aware? NI** Most scales used in this study can either be self-assessed or be assessed by an interviewer. The authors didn't provide details regarding who the assessors were. **Assessment have been influenced by knowledge of intervention? PY** **Assessment likely influenced by knowledge of intervention? PY** No matter whether the scales were assessed by the participants or the researchers, they were all affected by the knowledge of assigned intervention.	**Conclusion:** High risk **Selected from multiple eligible outcomes? Y** The authors used NPI, which produces two scores, the scores of distress and severity. Both scores were used in this study, without explanation of why only one of the scores were reported. **Selected from multiple eligible analyses? NI** The analysis intentions or plan were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened. **Selected from multiple eligible analyses? NI** The analysis intentions or plan were not reported. There is no study protocol or analysis plan available. Thus, we don't know whether the selection happened.	High risk

Appendix 3.7 Zarepour 2020

Domain 1: Randomisation process	Domain 2: Deviation from intended interventions	Domain 3: Blinding of participants & personnel	Domain 4: Blinding of outcome assessment	Domain 5: Selective reporting	Overall
Conclusion: Low risk **Allocation sequence concealment? Y** **Randomisation? Y** The study was randomised using “block randomization with a random sequence of 2 block sizes,” via a “block design software.” Thus, we can consider it a random, concealed. **Baseline differences? NI** The baselines for anxiety score were not significantly different. For other metrics, block randomisation was used to minimise the differences during randomisation. However, more data, such as age, would be helpful to validate the claim.	**Conclusion:** Low risk **Participants aware? Y** The authors said that “The caregivers signed written informed consents. Besides, they received verbal and written information about the study objectives, design, and duration.” Thus, when the participants were aware when they received the intervention. **Personnel aware? Y** The control group did not receive any formal education, as compared with the intervention group. Thus, the interventionists, who were supposed to deliver the education, were aware of the intervention. **Deviation from intervention due to trial context? PN** The authors mentioned that “the stabilizing of information provided by the researcher to the family caregivers of the intervention group during the 4-week training sessions [..]” As correction for deviation part of the protocol, it is reasonable to believe that the deviations were minimised. **Appropriate analysis to estimate the effects? PY** The study used ITT analysis and reported no subjects with missing data. Thus, the authors didn’t need to exclude any participants in the analysis, and the method was appropriate.	**Conclusion:** Low risk **Outcome data for all participants? Y** In Figure 1, the authors reported that the analysed sample size and the initial sample size were all 35 in each group. Thus, there were no subjects with missing data. Thus, Table 3 reported all the outcome data for all participants.	**Conclusion:** High risk **Inappropriate outcome measurement? N** “The psychometric properties of the inventory in Iranian population were confirmed by exploratory and confirmatory factor analysis. The correlation between the inventory and clinical specialists’ assessment also showed appropriate validity.19 The reliability of the Iranian form of the inventory was Cronbach’s alpha 0.88.2” Thus, the method was well-established in this population. **Measurement differed between groups? N** Both control and intervention groups used the same questionnaires. **Assessors aware? Y** The Spielberger anxiety questionnaire was completed by the caregivers themselves. Thus, the assessors knew the intervention. **Assessment have been influenced by knowledge of intervention? Y** Yes, the caregivers may undergo placebo effect. **Assessment likely influenced by knowledge of intervention? Y** Yes, Figure 2 shows worsening anxiety scores for the control group, suggesting for the nocebo effect on the control group.	**Conclusion:** Low risk **Selected from multiple eligible outcomes? N** The authors only measured the scores of anxiety and reported all scores of anxiety data. **Selected from multiple eligible analyses? N** Repeated measure ANOVA was a correct and also the most commonly used statistical analysis for the intended purpose. Thus, it is unlikely that the authors selectively reported the analyses. **Consistent with pre-specified analysis plan? PY** Although we didn’t find the pre-specified analysis plan for this study, the clinical trial protocol of this study can be found at https://www.irct.ir/trial/14691 The study reported its outcome as originally planned. Thus, the study was probably consistent with its analysis plan.	High risk

APPENDIX 4 DATA EXTRACTION FROM THE 2019 META-ANALYSIS

Setting up R environment

To replicate this study, you will need to install R, which is available at https://r-project.org/. For this preprocessing workflow, the current version of R on my computer is as follows:

```
R.Version()$version.string

## [1] "R version 4.2.2 (2022-10-31)"
```

To proceed, you must install and load the required packages, which is as follows:

```
# installation
install.packages("openxlsx")
install.packages("naniar")

# loading
library(openxlsx)
library(naniar)
```

Loading the dataset

Use the following code to load the dataset:

```
# load the dataset
dataset <- read.xlsx("Dataset.xlsx")

# replace the string value "-" with NA
dataset[dataset=="-"]  <- NA

# remove the rows all data item is NA
dataset <- dataset[rowSums(is.na(dataset)) != ncol(dataset), ]

# remove the columns all data item is NA
dataset <- dataset[,colSums(is.na(dataset))<nrow(dataset)]
```

When loading the dataset, R will alter some of the row names in case of incompatibility. After loading the dataset, we will first have a look at its size, with the following code:

```
# show the size of the dataset
dim(dataset)

## [1]  733 3705
```

This is a very large dataset with 733 rows and 3705 columns. This means the dataset contains 733 studies. We can first have a look at the first five rows of the dataset, with the following code to know what the columns might be about. The output of the code is omitted, as they are too long, due to the very large number of columns.

```
# show first five rows
head(dataset)
```

The 3705 columns are not all useful to us, except for the first several columns containing basic information. Because it will be very difficult for us to read all the columns with either Excel or R, we can first output all these names to a seperate file with the following code:

```
sink("colnames.txt")
colnames(dataset)
sink()
```

Screening by intervention type

Here we aim to filter the studies by the intervention type. First we can have a look at the column names related to “Intervention”:

```
df_cols <- colnames(dataset)
df_cols[grep('Intervention', df_cols)]

##  [1] "Intervention"
##  [2] "Intervention.types"
##  [3] "Intervention.subgroups"
##  [4] "Intervention.Brief.description.of.intervention"
##  [5] "Intervention.Brief.description.ofi.ntervention"
##  [6] "Intervention.Brief.intervention.description"
##  [7] "Intervention.Brief.introduction.to.the.intervention"
##  [8] "Intervention.Intervention.details"
##  [9] "Intervention.Intervention.details.1"
## [10] "Intervention.Intervention.details.(duration.and.intensity.of.the.intervention
,.dosage.of.drugs,.existence.of.a.protocol.or.manual.for.psychosocial,.training.or.edu
cation.interventions)"
## [11] "Intervention.intervention.type"
## [12] "Intervention.Intervention.type.1"
## [13] "Intervention.Interv..Type:"
## [14] "Intervention.Interv..type.[SELECT:.Pharmaceutical;.Traditional.Chinese.Medici
ne;.Other.trad..medicines;.Non-pharmacol.;.Multicomponent.(record.each.component);.Int
erv..for.carers;.End.of.life.care;.Treatment/prevention.of.co-morbidities;.Technology;
.Diagn"
## [15] "Intervention.Interv..type.[SELECT:.Pharmaceutical;.Traditional.Chinese.Medici
ne;.Other.trad..medicines;.Non-pharmacol.;.Multicomponent.(record.each.component);.Int
erv..for.carers;.End.of.life.care;.Treatment/preventioo-morbidities;.Technology;.Diagn
```

```
ostic;"
## [16] "Intervention.Name.of.intervention"
## [17] "Zarit.Burden.Interview.(ZBI).Post.Intervention.Change.mean"
## [18] "Zarit.Burden.Interview.(ZBI).Post.Intervention.Change.N
## [19] "Zarit.Burden.Interview.(ZBI).Post.Intervention.Change.SD"
```

Among these variables, "Intervention types" seems to provide most relevant information for fast screening:

```
knitr::kable(table(dataset$'Intervention.types'))
```

Var1	Freq
Comprehensive care interventions	7
Electrical brain stimulation	14
Multicomponent interventions	28
Music interventions	8
No intervention	251
Nutrition	4
Other interventions	2
Pharmaceutical	130
Physical exercise	17
Structured therapeutic psychosocial interventions	38
Supplements	44
Support for carers	14
Supportive psychosocial interventions	8
Traditional Chinese Medicine	156
Training and education for carers	6
Treatment of co-morbidities or additional risks	4
Unstructured therapeutic interventions	2

Thus, we should select 'Training and education for carers', 'Support for carers' and 'Comprehensive care interventions' as our desired intervention types, as these are consistent with the intervention types of the 7 included studies from the 2022 search. Therefore, we can use the following code:

```
df_studies <- dataset[dataset$'Intervention.types' %in% c(
    "Training and education for carers",
    "Support for carers",
    "Comprehensive care interventions"), ]
dim(df_studies)

## [1]   27 3705
```

As a result, there are 27 studies selected. Then we can have a look at the further details of the intervention.

```
knitr::kable(table(df_studies$'Intervention.subgroups'))
```

Var1	Freq
Community-based interventions	1
Group support and therapy	7
Individual-based interventions	6
Information, education, and training to support carers	7
Training and education for professional carers	2
Training and education for unpaid carers	4

```
# df_studies
selected_studies <- unique(df_studies$Study.Identifier)
rownames(df_studies) <- NULL
knitr::kable(df_studies[c('Study.Identifier', 'Intervention.types', 'Intervention.subg
roups', 'Country')])
```

Study.Identifier	Intervention.types	Intervention.subgroups	Country
Amit 2008	Training and education for carers	Training and education for unpaid carers	India
Pahlavanzadeh 2010	Support for carers	Information, education, and training to support carers	Iran
Wang 2010b	Comprehensive care interventions	Individual-based interventions	China
ZHAO 2010	Comprehensive care interventions	Individual-based interventions	China
TAN 2010	Training and education for carers	Training and education for unpaid carers	China
SUN 2010	Comprehensive care interventions	Individual-based interventions	China
Guerra 2011	Training and education for carers	Training and education for unpaid carers	Peru
Wang 2012e	Support for carers	Group support and therapy	China
SU 2012	Comprehensive care interventions	Individual-based interventions	China
JIANG 2012	Comprehensive care interventions	Individual-based interventions	China
Aboulafia-Brakha 2014	Support for carers	Group support and therapy	Brazil
Aboulafia-Brakha 2014	Support for carers	Information, education, and training to support carers	Brazil
Wang 2014a	Comprehensive care interventions	Individual-based interventions	China
Mercedeh 2014	Support for carers	Group support and therapy	Iran
Kamkhagi 2015	Support for carers	Information, education, and training to support carers	Brazil
Kamkhagi 2015	Support for carers	Group support and therapy	Brazil
Söylemez 2016	Support for carers	Information, education, and training to support carers	Turkey
Yang 2017a	Comprehensive care interventions	Community-based interventions	China
Wang 2017e	Training and education for carers	Training and education for professional carers	China
Wang 2017c	Training and education for carers	Training and education for professional carers	China
Shata 2017	Support for carers	Group support and therapy	Egypt
Salamizadeh 2017	Support for carers	Information, education, and training to support carers	Iran
Mahdavi 2017	Support for carers	Group support and therapy	Iran
Mahdavi 2017	Support for carers	Group support and therapy	Iran
Lök 2017	Support for carers	Information, education, and training to support carers	Turkey
Lin 2017	Training and education for carers	Training and education for unpaid carers	China
Orawan 2018	Support for carers	Information, education, and training to support carers	Thailand

Keyword screening (optional)

In this case, if the outcome measures of interest are known, we can do a keyword screening for relevant columns. For example, if we are interested in caregiver burden only, we can use the following code:

```
caregiver_burden_keys <- unique(c(
    grep("Burden", df_cols, ignore.case=TRUE),
    grep("ZBI", df_cols, ignore.case=TRUE),
    grep("Zarit", df_cols, ignore.case=TRUE)
    ))
caregiver_burden_keys <- df_cols[caregiver_burden_keys]

# show the number of selected columns
length(caregiver_burden_keys)

## [1] 91
```

Thus, we can find 91 relevant columns. Then we can subset the dataset according to our selection of studies and columns. For example, for the study that directly measures caregiver burden, we can use the following code:

```
# subset by caregiver burden-related columns
df_caregiver_burden <-
  dataset[which(rowSums(is.na(dataset[caregiver_burden_keys]))
                != ncol(dataset[caregiver_burden_keys])), ]
# subset by selected studies
df_caregiver_burden <-
  df_caregiver_burden[df_caregiver_burden$'Study.Identifier'
                      %in% as.list(selected_studies_list), ]

# drop rows and columns with only NA
df_caregiver_burden <-
  df_caregiver_burden[rowSums(is.na(df_caregiver_burden))
                      != ncol(df_caregiver_burden), ]
df_caregiver_burden <-
  df_caregiver_burden[,colSums(is.na(df_caregiver_burden))
                      < nrow(df_caregiver_burden)]

# show the size of simplified dataset
dim(df_caregiver_burden)

## [1]  20 152
```

Thus, we can generate a subset containing 20 studies with 152 non-empty columns and then save the subset to an Excel file with the following code:

```
# create the workbook
wb <- createWorkbook()

# add the worksheets
addWorksheet(wb, "caregiver_Burden")

# write the workbook
writeData(wb, "caregiver_Burden", df_caregiver_burden)

# save the workbook
saveWorkbook(wb, "Dataset_data_filtered.xlsx", overwrite = TRUE)
```

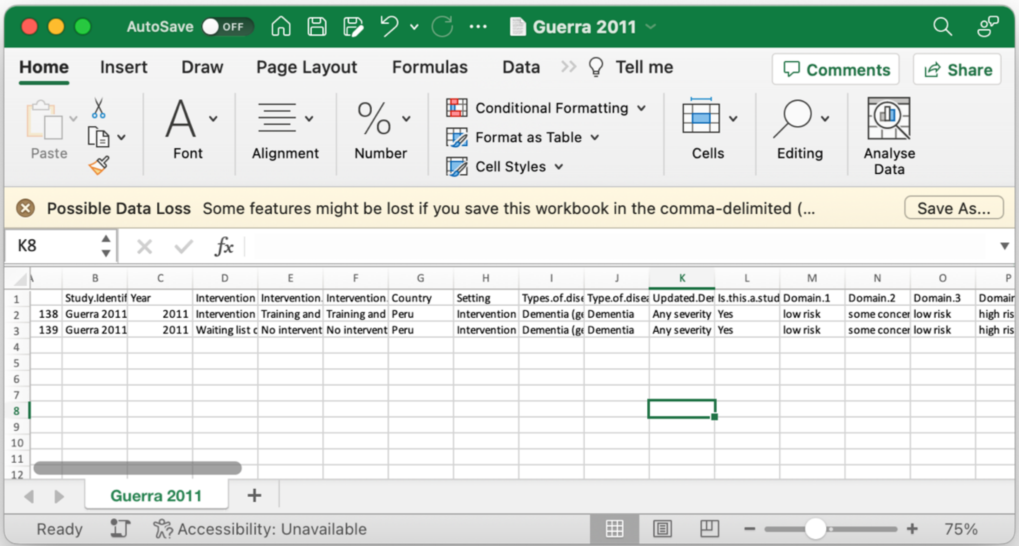
Then we can manually curate the dataset, with the Excel file that has reduced dimensions.

Extraction data from individual studies

In many cases, we are not certain of what effect measures the authors used, or maybe we just don't want to miss any effect measures that escape keyword screening. As the columns are especially redundant for individual studies, which means the data entry for each study will only use a very limited set of columns. Thus, extracting data from individual studies one by one can reduce the redundancy. One can do so with the following code:

```
# subset by selected studies
df_selected <-
  dataset[dataset$'Study.Identifier' %in%
            as.list(selected_studies_list), ]
df_selected <- df_selected[reserved_cols]

paste(c('study', '.csv'))

## [1] "study" ".csv"

for (study in selected_studies_list){
  study_file <- paste(study, '.csv')
  study_df <- dataset[dataset$Study.Identifier == study, ]
  study_df <- study_df[,colSums(is.na(study_df)) < nrow(study_df)]
  write.csv(study_df, study_file)
}
```

Thus, in the folder where the dataset is originally located, there will be 27 csv files named after each individual studies included. Each of them will be manually curated into a large dataset for meta-analysis. For example, the following screenshot of Guerra 2020.csv extracted from the original dataset only contains 3 rows and 61 columns, which is much more human-readable when compared with the original dataset which has over 3000 columns.



Apart from the basic information of the intervention, one may notice relevant outcomes of interests in the columns such as “Caregiver.distress.from.BPSD.(NPI-Q).6.months.N”. These data, from each of the 27 individual studies, can be finally curated into a large dataset that can be used for meta-analysis.

APPENDIX 5 META-ANALYSIS WITH R

Setting up R environment

To replicate this study, you will need to install R, which is available at https://r-project.org/. For this preprocessing workflow, the current version of R on my computer is as follows:

```
R.Version()$version.string

## [1] "R version 4.2.2 (2022-10-31)"
```

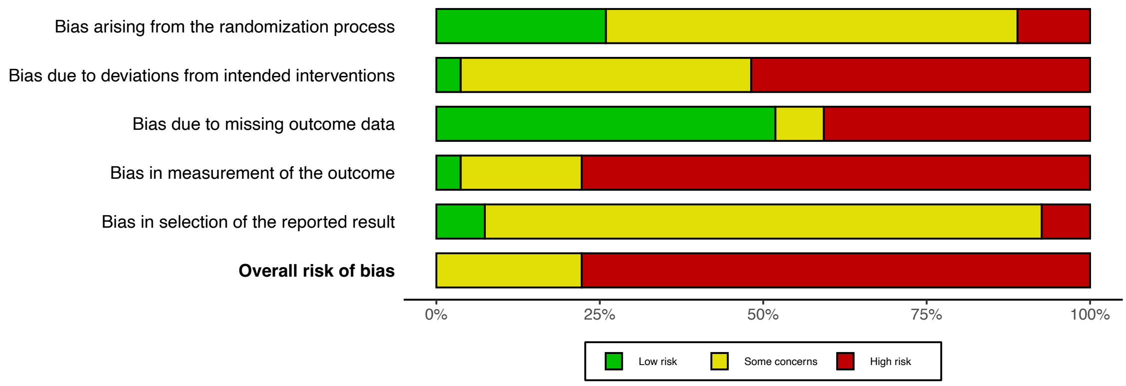
To proceed, you must install and load the required packages, which is as follows:

```
# installation
install.packages("openxlsx")
install.packages("metafor")
install.packages("robvis")


# loading
library(openxlsx)
library(metafor)
library(robvis)
```

Visualising risk of bias

```
# Loading the dataset
Dataset <- read.xlsx('Dataset final (RoB mod).xlsx', 'Combined')
RoB_cols <- c('Study.Identifier','Domain.1','Domain.2',
              'Domain.3','Domain.4','Domain.5','Overall')

# Remove duplicates
RoB_data <- unique(Dataset[RoB_cols])

# Show first 6 rows
head(RoB_data)

##      Study.Identifier      Domain.1      Domain.2  Domain.3  Domain.4
## 1           Amit 2008      low risk some concerns  low risk high risk
## 3           Chen 2020      low risk Some concerns  low risk high risk
## 9       Ghaffari 2019     high risk     high risk high risk high risk
## 15 Govindakumari 2020 Some concerns     high risk high risk high risk
## 19         Guerra 2011      low risk some concerns  low risk high risk
## 37         Hinton 2020      low risk      low risk  low risk  low risk
##         Domain.5   Overall
## 1  some concerns high risk
## 3  some concerns high risk
## 9  some concerns high risk
## 15 some concerns high risk
```

```
## 19 some concerns high risk
## 37      low risk  low risk

# visualise risk of bias in barplot
rob_summary(RoB_data, tool = "ROB2", weighted = FALSE)
```



Note that the output on R markdown may differ from the pdf output file. Thus, we use the following code to save the visualisation as pdf files:

```
# visualise risk of bias in barplot and save it as pdf file
pdf('RoB_barplot.pdf', width = 9, height = 3.2)
rob_summary(RoB_data, tool = "ROB2", weighted = FALSE)
dev.off()
```

Similarly, we can generate traffic light plot used in this article with the following code:

```
# subset the dataset by the source of data entries
rob_data.2022 <- unique(Dataset[Dataset$'Source' == '2022 search', ][RoB_cols])
rob_data.2019 <- unique(Dataset[Dataset$'Source' != '2022 search', ][RoB_cols])


# visualise risk of bias in traffic-light plot and save it as pdf file
pdf('RoB_taffic_2019.pdf', width = 7, height = 9)
rob_traffic_light(rob_data.2019, tool = "ROB2", psize = 5)
dev.off()

pdf('RoB_taffic_2022.pdf', width = 7, height = 4.5)
rob_traffic_light(rob_data.2022, tool = "ROB2", psize = 5)
dev.off()
```